

CHAPTER C26

WIND LOADS: GENERAL REQUIREMENTS

C26.1 PROCEDURES

Chapter 26 is the first of six chapters devoted to the wind load provisions. It provides the basic wind design parameters that are applicable to the various wind load determination methodologies contained in Chapters 27 through 31. Specific items covered in Chapter 26 include definitions, basic wind speed, exposure categories, internal pressures, elevation effects, enclosure classification, gust effects, and topographic factors.

C26.1.1 Scope. The procedures specified in this standard provide wind pressures and forces for the design of the main wind force resisting system (MWFRS) and of components and cladding (C&C) of buildings and other structures. The procedures involve the determination of wind directionality and velocity pressure, the selection or determination of an appropriate gust-effect factor, and the selection of appropriate pressure or force coefficients. The procedures account for the level of structural reliability required, the effects of differing wind exposures, the speed-up effects of certain topographic features such as hills and escarpments, and the size and geometry of the building or other structure under consideration. The procedures differentiate between rigid and flexible buildings and other structures, and the results generally envelope the most critical load conditions for the design of MWFRS as well as C&C.

The pressure and force coefficients provided in Chapters 27, 28, 29, and 30 have been assembled from the latest boundary-layer wind tunnel and full-scale tests and from previously available literature. Because the boundary-layer wind tunnel results were obtained for specific types of building, such as low- or high-rise buildings and buildings that have specific types of structural framing systems, the designer is cautioned against indiscriminate interchange of values among the figures and tables.

C26.1.2 Permitted Procedures. The wind load provisions provide several procedures (as illustrated in Fig. 26.1-1) from which the designer can choose.

For MWFRS:

1. Directional Procedure for Buildings of All Heights (Chapter 27).
2. Envelope Procedure for Low-Rise Buildings (Chapter 28).
3. Directional Procedure for Building Appurtenances and Other Structures (Chapter 29).
4. Wind Tunnel Procedure for All Buildings and Other Structures (Chapter 31).

For C&C:

1. Analytical Procedure for Buildings and Building Appurtenances (Chapter 30).
2. Wind Tunnel Procedure for All Buildings and Other Structures (Chapter 31).

A “simplified method” for which the designer can select wind pressures directly from a table without any calculation, where the building meets all the requirements for application of the method, is provided for designing buildings using the Directional Procedure (Chapter 27, Part 2), the Envelope Procedure (Chapter 28, Part 2), and the Analytical Procedure for Components and Cladding (Chapter 30).

Limitations. The provisions given under Section 26.1.2 apply to the majority of site locations and buildings and other structures, but for some projects, these provisions may be inadequate. Examples of site locations and buildings and other structures (or portions thereof) that may require other approved standards, special studies using applicable recognized literature pertaining to wind effects, or using the Wind Tunnel Procedure of Chapter 31, include

1. Site locations that have channeling effects or wakes from upwind obstructions. Channeling effects can be caused by topographic features (e.g., a mountain gorge) or buildings (e.g., a neighboring tall building or a cluster of tall buildings). Wakes can be caused by hills, buildings, or other structures.
2. Buildings with unusual or irregular geometric shape, including barrel vaults, arched roofs, and other buildings whose shape (in plan or vertical cross section) differs significantly from the shapes in Figs. 27.3-1, 27.3-2, 27.3-3, 27.3-7, 28.3-1, and 30.3-1 through 30.3-7. Unusual or irregular geometric shapes include buildings with multiple setbacks, curved facades, or irregular plans resulting from significant indentations or projections, openings through the building, or multitower buildings connected by bridges.
3. Buildings or other structures with response characteristics that result in substantial vortex-induced and/or torsional dynamic effects, or dynamic effects resulting from aeroelastic instabilities such as flutter or galloping. Such dynamic effects are difficult to anticipate, being dependent on many factors, but should be considered when any one or more of the following apply:
 - The height of the building or other structure is more than 400 ft (122 m).
 - The height of the building or other structure is greater than 4 times its minimum effective width $B_{\min}$, as defined below.
 - The lowest natural frequency of the building or other structure is less than $n_1 = 0.25$ Hz.
 - The reduced velocity $\bar{V}_z / (n_1 B_{\min}) > 5$, where $\bar{z} = 0.6h$ and $\bar{V}_z$ is the mean hourly velocity in ft/s (m/s) at height $\bar{z}$.

The minimum effective width $B_{\min}$ is defined as the minimum value of $\sum h_i B_i / \sum h_i$ considering all wind

directions. The summations are performed over the height of the building or other structure for each wind direction, where h_i is the height above grade of level i and B_i is the width at level i normal to the wind direction.

4. Bridges, cranes, electrical transmission lines, guyed masts, highway signs and lighting structures, telecommunication towers, and flagpoles.

When undertaking detailed studies of the dynamic response to wind forces, the fundamental frequencies of the building or other structure in each direction under consideration should be established using the structural properties and deformational characteristics of the resisting elements in a properly substantiated analysis, and not using approximate equations based on height.

Shielding. Because of the lack of reliable analytical procedures for predicting the effects of shielding provided by buildings and other structures or by topographic features, reductions in velocity pressure caused by shielding are not permitted under the provisions of this chapter. However, this does not preclude the determination of shielding effects and the corresponding reductions in velocity pressure by means of the Wind Tunnel Procedure in Chapter 31.

C26.2 DEFINITIONS

Several important definitions given in the standard are discussed in the following text. These terms are used throughout the standard and are provided to clarify application of the standard provisions.

BUILDING, ENCLOSED; BUILDING, OPEN; BUILDING, PARTIALLY ENCLOSED; BUILDING, PARTIALLY OPEN: These definitions relate to the proper selection of internal pressure coefficients, (GC_{pi}). “Enclosed,” “open,” and “partially enclosed” buildings are specifically defined. All other buildings are considered to be “partially open” by definition, although there may be large openings in two or more walls. An example of this would be a parking garage through which the wind can easily pass but which does not meet the definition for either an open or a partially enclosed building. The internal pressure coefficient for such a building would be ± 0.18 , and the internal pressures would act on the solid areas of the walls and roof. The standard also specifies that a building that meets both the “open” and “partially enclosed” definitions should be considered “open.”

BUILDING OR OTHER STRUCTURE, FLEXIBLE: A building or other structure is considered “flexible” if it contains a significant dynamic resonant response. Resonant response depends on the gust structure contained in the approaching wind, on wind loading pressures generated by the wind flow about the building, and on the dynamic properties of the building or structure. Gust energy in the wind is smaller at frequencies above about 1 Hz. Therefore, the resonant response of most buildings and structures with lowest natural frequency above 1 Hz are sufficiently small that resonant response can often be ignored. The natural frequency of buildings or other structures greater than 60 ft (18.3 m) in height is determined in accordance with Sections 26.11.1 and 26.11.2. When buildings or other structures have a height exceeding 4 times the least horizontal dimension or when there is reason to believe that the natural frequency is less than 1 Hz (natural period greater than 1 s), the natural frequency of the structure should be investigated. Approximate equations for natural frequency or period for various building and structure types in addition to those given in Section 26.11.2 for buildings are contained in Commentary Section C26.11.

BUILDING OR OTHER STRUCTURE, REGULAR-SHAPED: Defining the limits of applicability of the various procedures within the standard requires a balance between the practical need to use the provisions past the range for which data have been obtained and restricting use of the provisions past the range of realistic application. Wind load provisions are based primarily on wind tunnel tests on shapes shown in Figs. 27.3-1, 27.3-2, 27.3-3, 27.3-7, 28.3-1, and 30.3-1 through 30.3-7. Extensive wind tunnel tests on actual structures under design show that relatively large changes from these shapes can, in many cases, have minor changes in wind load, while in other cases seemingly small changes can have relatively large effects, particularly on cladding pressures. Wind loads on complicated shapes are frequently smaller than those on the simpler shapes of Figs. 27.3-1, 27.3-2, 27.3-7, 28.3-1, and 30.3-1 through 30.3-7, and so wind loads determined from these provisions are expected to envelop most structure shapes. Buildings or other structures that are clearly unusual should be designed using the Wind Tunnel Procedure of Chapter 31.

BUILDING OR OTHER STRUCTURE, RIGID: The defining criterion for “rigid,” in comparison to “flexible,” is that the natural frequency is greater than or equal to 1 Hz. A general guidance is that most rigid buildings and structures have height-to-minimum-width less than 4. The provisions of Sections 26.11.1 and 26.11.2 provide methods for calculating natural frequency (period = 1/natural frequency), and Commentary Section C26.11 provides additional guidance.

COMPONENTS AND CLADDING (C&C): Components receive wind loads directly or from cladding and transfer the load to the MWFRS. Cladding receives wind loads directly. Examples of components include, but are not limited to, fasteners, purlins, girts, studs, sheathing, roof decking, certain trusses, and elements of trusses receiving wind loads from cladding. Examples of cladding include, but are not limited to, wall coverings, curtain walls, roof coverings, sheathing, roof decking, exterior windows, and doors. Components can be part of the MWFRS when they act as elements in shear walls or roof diaphragms, but they may also be loaded directly by wind as individual elements. The designer should use appropriate loads for design of components, which may require certain components to be designed for more than one type of wind loading; for example, long-span roof trusses should be designed for loads associated with MWFRS, and individual members of trusses should also be designed for C&C loads (Mehta and Marshall 1998).

DIAPHRAGM: This definition for diaphragm in wind load applications, for the case of untopped steel decks, differs somewhat from the definition used in Section 12.3 because diaphragms under wind loads are expected to remain essentially elastic.

EFFECTIVE WIND AREA, A : Effective wind area is the area of the building surface used to determine (GC_p). This area does not necessarily correspond to the area of the building surface contributing to the force being considered. Two cases arise. In the usual case, the effective wind area does correspond to the area tributary to the force component being considered. For example, for a cladding panel, the effective wind area may be equal to the total area of the panel. For a cladding fastener, the effective wind area is the area of cladding secured by a single fastener. A mullion may receive wind from several cladding panels. In this case, the effective wind area is the area associated with the wind load that is transferred to the mullion.

The second case arises where components such as roofing panels, wall studs, or roof trusses are spaced closely together. The area served by the component may become long and narrow. To better approximate the actual load distribution in such cases,

the width of the effective wind area used to evaluate (GC_p) need not be taken as less than one-third the length of the area. This increase in effective wind area has the effect of reducing the average wind pressure acting on the component. Note, however, that this effective wind area should only be used in determining the (GC_p) in Figs. 30.3-1 through 30.3-6. The induced wind load should be applied over the actual area tributary to the component being considered.

For membrane roof systems, the effective wind area is the area of an insulation board (or deck panel if insulation is not used) if the boards are fully adhered (or the membrane is adhered directly to the deck). If the insulation boards or membrane are mechanically attached or partially adhered, the effective wind area is the area of the board or membrane secured by a single fastener or individual spot or row of adhesive.

For windows, doors, and other fenestration assemblies, the effective wind area for typical single-unit assemblies can be taken as the overall area of the assembly. For assemblies comprised of more than one unit mulled together or for more complex fenestration systems, it is recommended that the fenestration product manufacturer be consulted for guidance on the appropriate effective wind area to use when calculating the design wind pressure for product specification purposes.

The definition of effective wind area for rooftop solar panels and arrays is similar to that for components and cladding. As with C&C, the width of the effective wind area need not be less than one-third its length (which is typically equal to the span of the framing element being considered). The induced wind pressure is calculated per Fig. 29.4-4 using this effective wind area, and the wind pressure is then applied over the actual area tributary to the element.

Effective wind area is equal to the tributary area except in cases where the exception is invoked that the width of the effective wind area need not be less than one-third its length. In such cases, the effective wind area can be taken as larger than the tributary area.

Tributary area for a spanning structural member of a solar array depends on the span length of that member times the perpendicular distances to adjacent parallel members. For a support point or fastener, tributary area depends on the span of members framing into that support point.

Tributary area (and effective wind area) can depend on the characteristics of the solar array support system and the load path.

For a roof-bearing system that has different load paths for upward, downward, and lateral forces, the appropriate effective wind area for each direction of forces is used.

If the support system for the solar array has adequate strength, stiffness, and interconnectedness to span across a support or ballast point that is subject to yielding or uplift, the effective wind area can be correspondingly increased, provided that strengths are not governed by brittle failure and that the deformation of the array is evaluated and does not result in adverse performance. It should be noted that effective wind areas for uplift are usually much smaller than for lateral (drag) forces for ballasted arrays.

MAIN WIND FORCE RESISTING SYSTEM (MWFRS):

The MWFRS can consist of a structural frame or an assemblage of structural elements that work together to transfer wind loads acting on the entire building or structure to the ground. Structural elements such as cross-bracing, shear walls, roof trusses, and roof diaphragms are part of the MWFRS when they assist in transferring overall loads (Mehta and Marshall 1998).

WIND-BORNE DEBRIS REGIONS: Wind-borne debris regions are defined to alert the designer to areas requiring consideration of missile impact design. These areas are located within hurricane-prone regions where there is a high risk of glazing failure caused by the impact of wind-borne debris.

C26.3 SYMBOLS

The following additional symbols and notation are used herein:

A_{ob} = average area of open ground surrounding each obstruction;

n = reference period, in years;

P_a = annual probability of wind speed exceeding a given magnitude [Eq. (C26.5-3)];

P_n = probability of exceeding design wind speed during n years [Eq. (C26.5-3)];

S_{ob} = average frontal area presented to the wind by each obstruction;

V_t = wind speed averaged over t seconds (see Fig. C26.5-1), in mi/h (m/s);

V_{3600} = mean wind speed averaged over 1 h (see Fig. C26.5-1), in mi/h (m/s); and

β = damping ratio (percentage of critical damping).

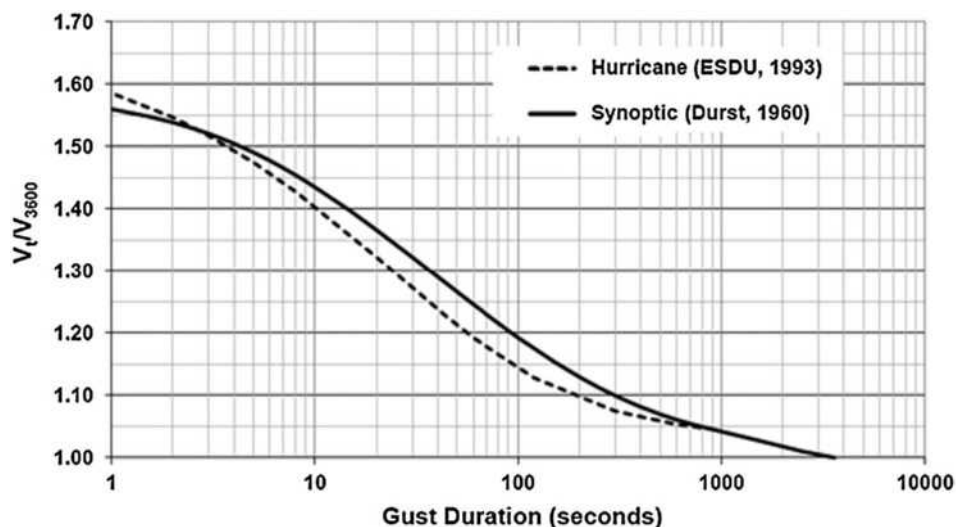


FIGURE C26.5-1 Maximum Speed Averaged over t (s) to Hourly Mean Speed

C26.4 GENERAL

C26.4.3 Wind Pressures Acting on Opposite Faces of Each Building Surface. Section 26.4.3 is included in the standard to ensure that internal and external pressures acting on a building surface are taken into account by determining a net pressure from the algebraic sum of those pressures. For additional information on the application of the net C&C wind pressure acting across a multilayered building envelope system, including air-permeable cladding, refer to Section C30.1.1.

C26.5 WIND HAZARD MAP

C26.5.1 Basic Wind Speed. All the wind speed maps in ASCE 7-16 have been updated, based on (i) a new analysis of non-hurricane wind data available through 2010, and (ii) improvements to the hurricane simulation model, which better account for the translation speed effects of fast-moving storms and the transition from hurricanes to extratropical storms in the northern latitudes (i.e., transition from warm core to cold core low-pressure systems). Separate wind speed maps are now provided for Risk Category III and Risk Category IV buildings and structures, recognizing the higher reliabilities required for essential facilities and facilities whose failure could pose a substantial hazard to the community. Location-specific basic wind speeds may be determined using <http://windspeedatcouncil.org/windspeed>. This website provides wind speeds to the nearest mile per hour based on a defined location using either latitude/longitude or an address. The website results use the same data used to develop the paper maps currently in the standard. Wind speeds are provided to the user for each of the Risk Categories, each of the serviceability periods, and a comparison speed for ASCE 7-10, ASCE 7-05 (3-s peak gust), and ASCE 7-93 (fastest mile).

In the 2016 edition, microzoned “effective” wind speed maps for Hawaii were added in accordance with the strength design return periods, including the effect of topography. The Hawaii effective wind speeds are algebraically formulated to include the macroscale and mesoscale terrain-normalized values of K_{zt} and K_d (Chock et al. 2005), i.e., $V_{\text{effective}}$ is the basic windspeed V multiplied by $\sqrt{(K_{zt} \times K_d / 0.85)}$, so that the engineer is permitted to more conveniently use the standard values of K_{zt} of 1.0 and K_d as given in Table 26.6-1. Note that local site conditions of finer topoccale, such as ocean promontories and local escarpments, should still be examined. Spatial resolution scales for digital modeling, including terrain effects, are conventionally described in the recognized literature as follows:

Scale	Spatial Resolution
Toposcale	32–656 ft (10–200 m)
Mesoscale	656 ft–3.1 mi (200 m–5 km)
Macroscale	3.1 mi–311 mi (5–500 km)

The decision in ASCE 7-10 to move to separate strength design wind speed maps for different Risk Categories in conjunction with a wind load factor of 1.0, instead of using a single map in conjunction with an Importance Factor and a load factor of 1.6, relied on several considerations:

1. A strength-level design wind speed map brings the wind loading approach in line with that used for seismic loads, in that they both are aimed at achieving uniform risk rather than uniform hazard and eliminate the use of a load factor for strength design.
2. Having separate maps removes inconsistencies that occurred with the use of Importance Factors, which varied with location, and allows for the geographical description

of zones affected by nonhurricane winds only and by both hurricane and non-hurricane winds as a function of mean recurrence interval (MRI).

3. Each map has the same MRI for design wind speeds in those two zones.
4. By providing the design wind speed directly, the maps more clearly inform owners and their consultants about the storm intensities for which the buildings and other structures are designed.

Selection of Return Periods. The methodology for selection of the return periods used in ASCE 7-10 (Vickery et al. 2010) has been modified for ASCE 7-16. In order to determine a return period for each Risk Category consistent with the target reliabilities in Table C1.3-1, the ASCE 7 Load Combinations Subcommittee conducted a reliability analysis that incorporated new data on the directionality factor. The nominal design value $K_d = 0.85$ was based on a relatively simple directional analysis conducted as part of the original ANSI A58/ASCE 7 load factor development. One of the underlying assumptions of the original analysis was that the wind directionality factor, $K_d = 0.85$, was unbiased because only limited data on the effects of wind directionality were available at the time. More recent research by Isyumov et al. (2013), simulating three building geometries at two different locations, indicates that the ASCE 7 nominal values of K_d are affected by a bias (defined as the ratio of the mean value, μK_d , to the nominal value, K_{dn}). The nominal value of K_d is conservative for both tropical and extratropical winds because the mean value is less than the nominal value. Additional reliability analyses were performed to examine the effect of K_d on the return period and associated reliability. The subcommittee found that the following return periods for each Risk Category are consistent with the target reliabilities in the first row of Table 1.3-1: Risk Category I: 300 years; Risk Category II: 700 years; Risk Category III: 1,700 years; and Risk Category IV: 3,000 years.

Wind Speed. The wind speed maps of Figs. 26.5-1 and 26.5-2 present basic wind speeds for the contiguous United States, Alaska, Hawaii, and other selected locations. The wind speeds correspond to 3-s gust speeds at 33 ft (10 m) above ground for Exposure Category C. Because the wind speeds of Figs. 26.5-1 and 26.5-2 reflect conditions at airports and similar open-country exposures, they do not account for the effects of significant topographic features such as those described in Section 26.8. In ASCE 7-16, wind speeds in nonhurricane-prone areas of the contiguous United States are mapped using contours to better reflect regional variations in the extreme wind climate. Point values are provided to aid interpolation, in a style similar to that used in the ASCE 7 seismic hazard maps. Summaries of the data and methods used to estimate both the nonhurricane and hurricane wind speeds are given below, along with a description of how these wind speeds are combined to make the final maps. Detailed descriptions are provided in Pintar et al. (2015) and Lombardo et al. (2016).

Nonhurricane Wind Speeds. The nonhurricane wind speeds for the contiguous United States were estimated from peak gust speed data collected at 575 meteorological stations. The data at each station were extracted from the meteorological records and classified by storm type, thunderstorm or nonthunderstorm after removal of gusts associated with tropical cyclones (i.e., hurricanes and tropical storms). Recorded peak gusts from each station were corrected as needed to standardize the observations to equivalent 3-s peak gusts at 10 m (33 ft) height over open (Exposure C) terrain. At each station, there were at least 15 years of data, and there were sufficient numbers of both thunderstorm and nonthunderstorm observations to account for their potential

differences when estimating wind speeds with specified mean recurrence intervals. The estimation was performed in two stages. In the first stage, a peaks-over-thresholds (POT) model was fitted to the data from each station. The POT model used was the Poisson process model first described in Pickands (1971) and extended in Smith (1989) to allow the parameters of the Poisson process to be time dependent. This model allowed for differentiation between thunderstorm and nonthunderstorm winds. The Poisson process has a tail length parameter that may be set to zero, leading to Gumbel-like tails for the distribution of wind speeds. Such distributional tails were used in this work, consistent with past practice in wind engineering. The fitted POT models allowed for the estimation of wind speeds for any required mean recurrence interval at all stations. In the second stage, local regression (Cleveland and Devlin 1988) was used to interpolate wind speeds at all points of a fine regular grid covering the contiguous United States for all required mean recurrence intervals. This had the effect of spatially smoothing the noisy station estimates. The smoothed wind speed estimates provided the basis for creating the isotach maps.

Limited data were available on the Washington and Oregon coast. In this region, a special wind region was defined to permit local jurisdictions to select speeds based on local knowledge and analysis. Speeds in the Aleutian Islands and in the interior of Alaska were established from gust data. Insufficient data were available for a detailed coverage of the mountainous regions, so gust data in Alaska were not corrected for potential terrain influence. It is possible that wind speeds in parts of Alaska would be smaller if the topographic wind speed-up effect on recorded wind speeds were taken into account. In Alaska, the maps for each return period were determined by multiplying the 50-year MRI contours given in ASCE 7-10 Fig. CC-3 by a factor, F_{RA} , equal to

$$F_{RA} = 0.45 + 0.085 \ln(12T) \quad (C26.5-1)$$

where T is the return period in years (Peterka and Shahid 1998). The resulting contours were interpolated to the nearest 10 mi/h, except for the innermost and outermost contours, which were rounded to the nearest 5 mi/h.

Hurricane Wind Speeds. The hurricane wind speeds are based on the results of a Monte Carlo simulation model generally described in Applied Research Associates (2001), Vickery and Wadhera (2008a, b), and Vickery et al. (2009a, b, 2010). The hurricane simulation model used to develop the wind speeds in ASCE 7-16 included two updates to the model used for ASCE 7-10. A reduced translation speed effect for fast-moving storms (NRC 2011) was incorporated, and a simple extratropical transition model was also implemented, where the surface winds are reduced linearly by up to 10% over the latitude range 37°N to 45°N. This reduction approximates transitioning from a hurricane boundary layer to an extratropical storm boundary layer. The effects of the model revisions are to slightly reduce hurricane speeds in the northeast, extending from Maine to Virginia.

Combination of Nonhurricane and Hurricane Wind Speed Data. Nonhurricane wind speeds and hurricane wind speeds were estimated for return periods ranging from 10 years to 100,000 years. The nonhurricane and hurricane winds were then combined as statistically independent events using Eq. (C26.5-2), the same general approach that has been used in previous editions of ASCE 7:

$$P_a(v > V) = 1 - P_{NH}(v < V)P_H(v < V) \quad (C26.5-2)$$

where

$P_a(v > V)$ is the annual exceedance probability for the combined wind hazards,

$P_{NH}(v < V)$ is the annual nonexceedance probability for nonhurricane winds, and

$P_H(v < V)$ is the annual nonexceedance probability for hurricane winds.

The combined winds were interpolated to yield the combined wind hazard curves for MRIs associated with each of the wind speed maps. In cases where the hurricane contours are unchanged from ASCE 7-10, the shape files from these previous maps were used to ensure continuity between the maps.

Correlation of Basic Wind Speed Map with the Saffir-Simpson Hurricane Wind Scale. Hurricane intensities are reported by the National Hurricane Center (NHC) according to the Saffir-Simpson Hurricane Wind Scale (NHC 2015), shown in Table C26.5-1. This scale has found broad usage by hurricane forecasters and local and federal agencies responsible for short-range evacuation of residents during hurricane alerts, as well as long-range disaster planners and the news media. The scale contains five categories of hurricanes and distinguishes them based on wind speed intensity.

The wind speeds used in the Saffir-Simpson Hurricane Wind Scale are defined in terms of a sustained wind speed with a 1-min averaging time at 33 ft (10 m) over open water. The ASCE 7 standard by comparison uses a 3-s gust speed at 33 ft (10 m) above ground in Exposure C (defined as the basic wind speed, and shown in the wind speed map, Figs. 26.5-1 and 26.5-2). The sustained wind speed over water in Table C26.5-2 cannot be converted to a peak gust wind speed using the Durst curve of Fig. C26.5-1, which is only valid for wind blowing over open terrain (Exposure C). An approximate relationship between the wind speeds in ASCE 7 and the Saffir-Simpson scale, based on recent data which indicate that the sea surface roughness remains approximately constant for mean hourly speeds in excess of 67 mi/h (30 m/s), is shown in Table C26.5-2. The table provides the sustained wind speeds of the Saffir-Simpson Hurricane Wind Scale over water, equivalent-intensity gust wind speeds over water, and equivalent-intensity gust wind speeds over land. For a storm of a given intensity, Table C26.5-2 takes into consideration both the reduction in wind speed as the storm moves from over water to over land because of changes in surface roughness, and the change in the gust factor as the storm moves from over water to over land (Vickery et al 2009a; Simiu et al. 2007).

Table C26.5-3 shows the design wind speed from the ASCE 7 basic wind speed maps (Figs. 26.5-1 and 26.5-2) for various

Table C26.5-1 Saffir-Simpson Hurricane Wind Scale

Hurricane Category	Sustained Wind Speed ^a mph (m/s)	Types of Damage Due to Hurricane Winds
1	74–95 (33–42)	Very dangerous winds will produce some damage
2	96–110 (43–49)	Extremely dangerous winds will cause extensive damage
3	111–129 (50–57)	Devastating damage will occur
4	130–156 (58–69)	Catastrophic damage will occur
5	≥157 (70)	Highly catastrophic damage will occur

^a1-minute average wind speed at 33 ft (10 m) above open water.

Table C26.5-2 Approximate Relationship between Wind Speeds in ASCE 7 and Saffir-Simpson Hurricane Wind Scale

Saffir-Simpson Hurricane Category	Sustained Wind Speed over Water ^a		Gust Wind Speed over Water ^b		Gust Wind Speed Over Land ^c	
	mph	m/s	mph	m/s	mph	m/s
1	74–95	33–42	90–116	40–51	81–105	36–47
2	96–110	43–49	117–134	52–59	106–121	48–54
3	111–129	50–57	135–157	60–70	122–142	55–63
4	130–156	58–69	158–190	71–84	143–172	64–76
5	>157	>70	>191	>85	>173	>77

^a1-min average wind speed at 33 ft (10 m) above open water.

^b3-s gust wind speed at 33 ft (10 m) above open water.

^c3-s gust wind speed at 33 ft (10 m) above open ground in Exposure Category C. This column has the same basis (averaging time, height, and exposure) as the basic wind speed from Figs. 26.5-1 and 26.5-2.

locations along the hurricane coastline from Maine to Texas, and for Hawaii, Puerto Rico, and the Virgin Islands. Tables C26.5-4 through C26.5-6 show the basic wind speeds for Risk Category II, III, and IV buildings and other structures in terms of the hurricane category equivalents on the Saffir-Simpson Hurricane Wind Scale. These wind speeds represent an approximate limit state; structures designed to withstand the wind loads specified in this standard, which are also appropriately constructed and maintained, should have a high probability of surviving hurricanes of the intensities shown in Tables C26.5-4 through C26.5-6 without serious structural damage from wind pressure alone.

Tables C26.5-2 through C26.5-6 are intended to help users of the standard better understand design wind speeds as used in this standard in relation to wind speeds reported by weather forecasters and the news media, who commonly use the Saffir-Simpson Hurricane Wind Scale. The Saffir-Simpson hurricane category equivalent Exposure C gust wind speed values given in Tables C26.5-2 through C26.5-6, which are associated with a given sustained wind speed, should be used as a guide only.

Table C26.5-3 Basic Wind Speeds at Selected Coastal Locations in Hurricane-Prone Areas

Location	Coordinates (decimal degrees)		Basic Wind Speeds (mph)		
	Latitude	Longitude	Risk Cat II (700-yr)	Risk Cat III (1,700-yr)	Risk Cat IV (3,000-yr)
Bar Harbor, Maine	44.3813	–68.1968	110	119	124
Hampton Beach, New Hampshire	42.9107	–70.8102	116	124	129
Boston, Massachusetts	42.3578	–71.0012	120	129	133
Hyannis, Massachusetts	41.6359	–70.2901	132	139	145
Newport, Rhode Island	41.453	–71.3058	131	139	144
New Haven, Connecticut	41.2803	–72.9327	121	129	134
Southampton, New York	40.871	–72.3844	130	138	145
Manhattan, New York	40.7005	–74.0135	115	127	130
Atlantic City, New Jersey	39.3536	–74.4336	126	135	140
Rehoboth Beach, Delaware	38.7167	–75.0752	121	130	137
Ocean City, Maryland	38.3314	–75.0835	128	136	140
Virginia Beach, Virginia	36.8306	–75.9691	123	132	137
Wrightsville Beach, North Carolina	34.1973	–77.8014	146	156	160
Folly Beach, South Carolina	32.6496	–79.9512	149	158	166
Sea Island, Georgia	31.179	–81.3472	131	145	150
Jacksonville Beach, Florida	30.2836	–81.387	129	140	150
Melbourne Beach, Florida	28.0684	–80.5564	151	162	170
Miami Beach, Florida	25.7643	–80.1309	170	183	190
Key West, Florida	24.5477	–81.7843	180	200	200
Clearwater, Florida	27.9658	–82.8042	146	153	158
Panama City Beach, Florida	30.1558	–85.7744	135	146	149
Gulf Shores, Alabama	30.2486	–87.6808	160	172	177
Biloxi, Mississippi	30.3924	–88.8887	160	176	183
Slidell, Louisiana	30.2174	–89.824	142	152	158
Cameron, Louisiana	29.7761	–93.2921	144	154	158
Galveston, Texas	29.2663	–94.826	150	159	167
Port Aransas, Texas	27.8346	–97.0446	150	157	163
Hawaii	n/a	n/a	130	145	150
San Juan, Puerto Rico	18.4501	–66.0367	160	170	175
Virgin Islands	n/a	n/a	165	175	180

Notes:

1. All wind speeds in Table C26.5-3 are 3-s gust wind speeds at 33 ft (10 m) above ground for Exposure Category C.
2. The basic wind speed in hurricane-prone regions can vary significantly over a city or county; the values shown are for randomly selected points along the coast for communities listed in the tables. Wind speeds at other locations within those communities may be greater or less than the values shown in the table.
3. Conversion of mph to m/s: $\text{mph} \times 0.44704 = \text{m/s}$.

Table C26.5-4 Basic Wind Speed for Risk Category II Buildings and Other Structures at Selected Locations in Hurricane-Prone Areas

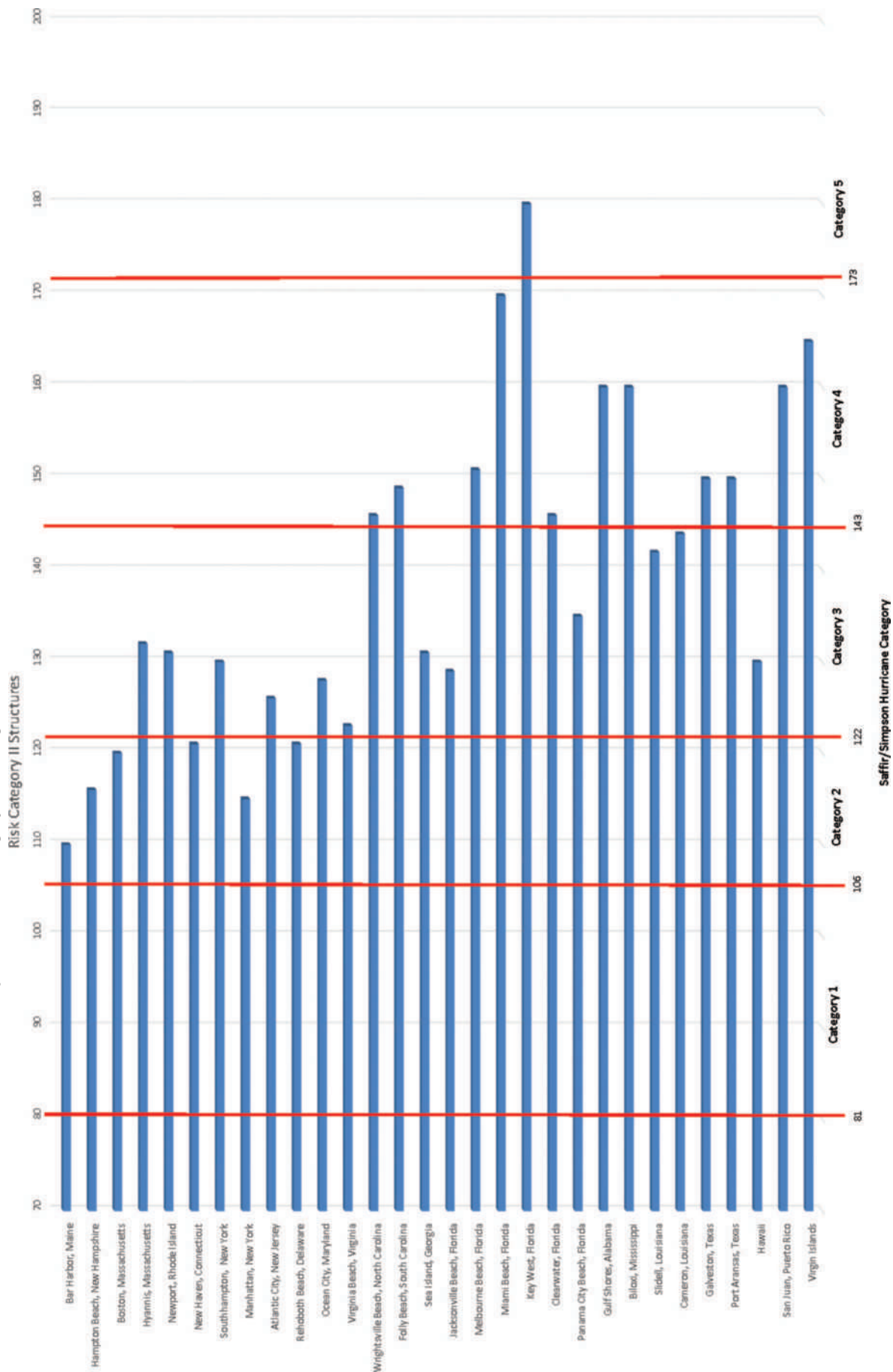


Table C26.5-5 Basic Wind Speed for Risk Category III Buildings and Other Structures at Selected Locations in Hurricane-Prone Areas

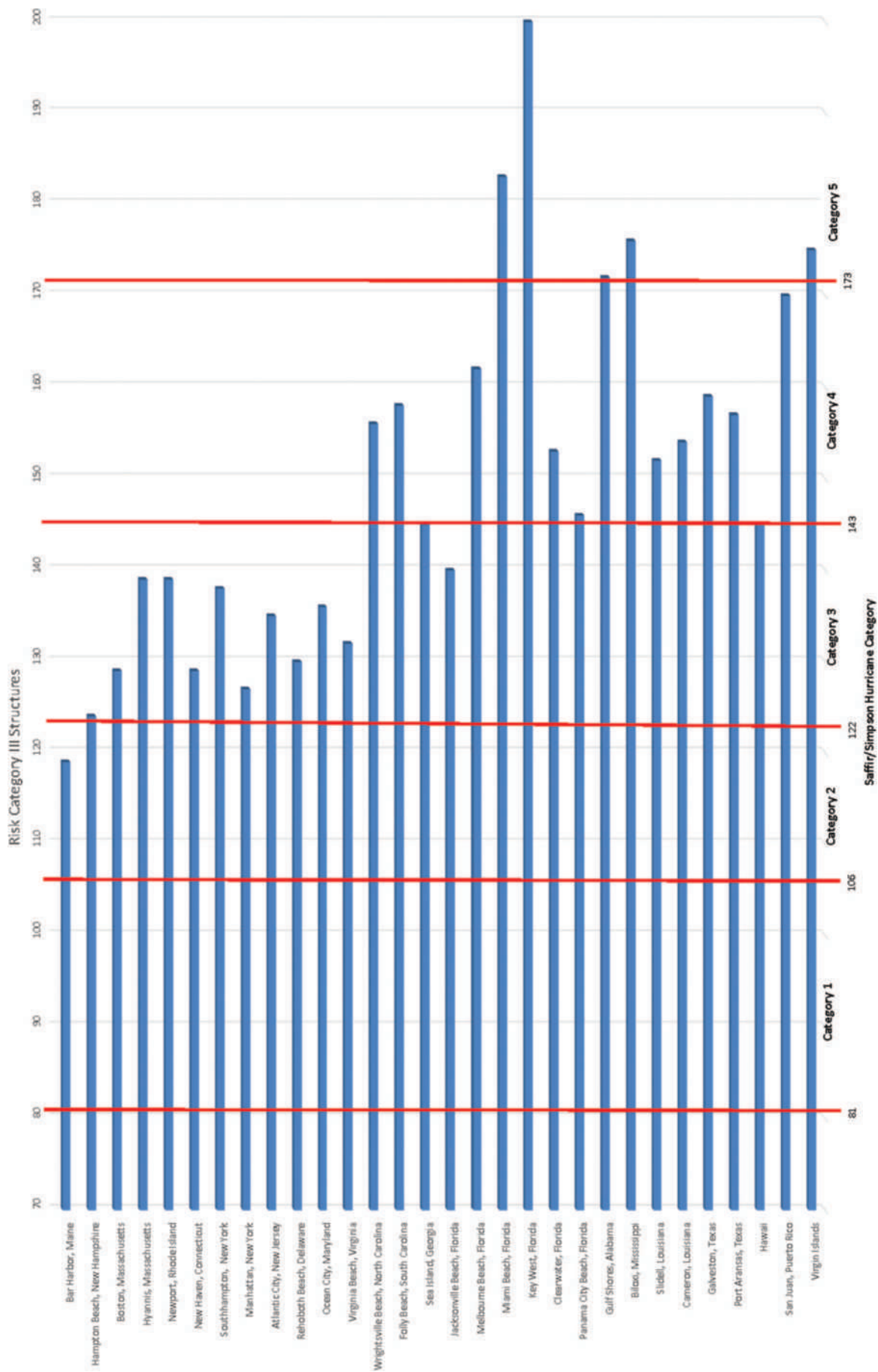
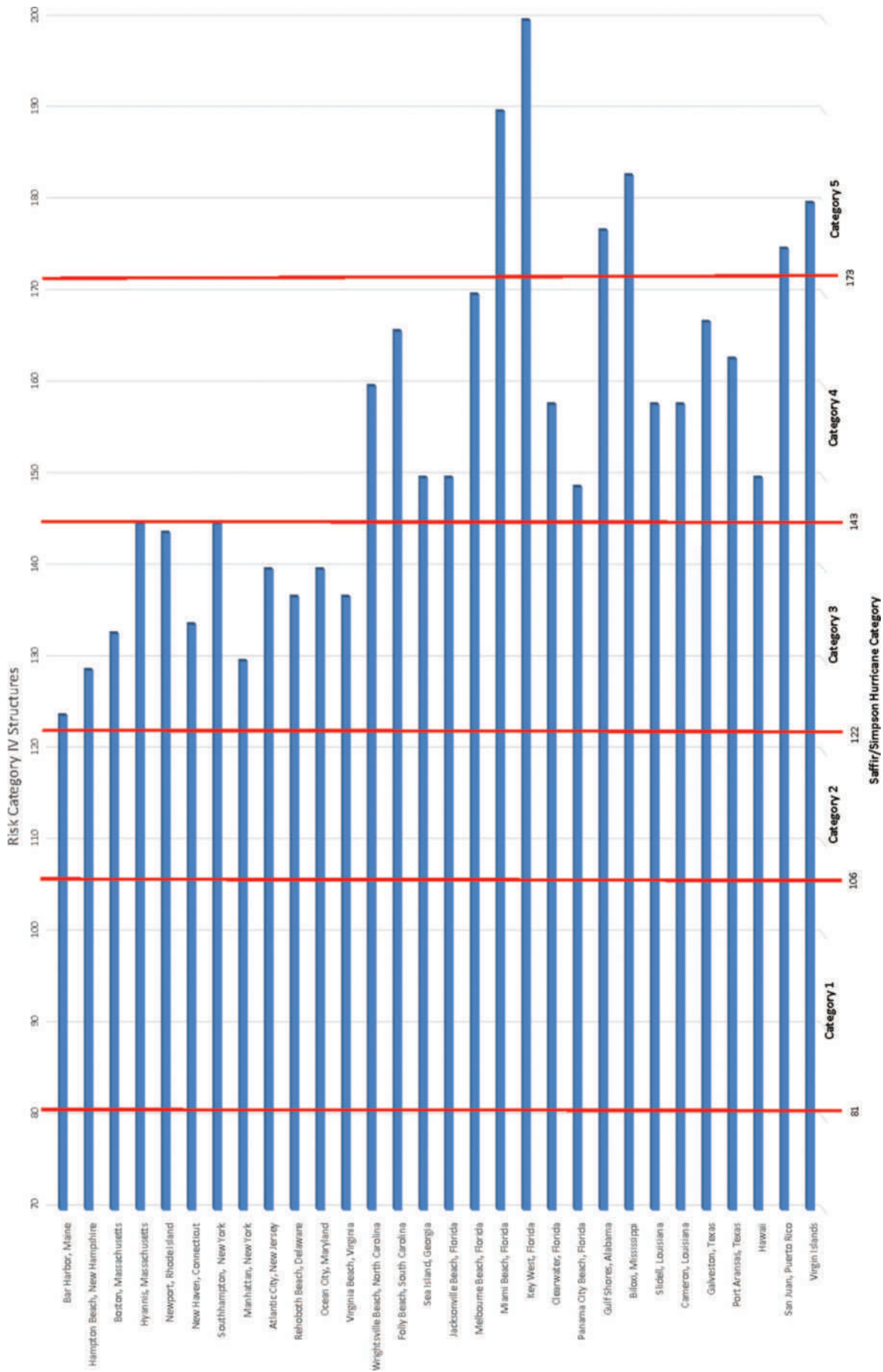


Table C26.5-6 Basic Wind Speed for Risk Category IV Buildings and Other Structures at Selected Locations in Hurricane-Prone Areas



The gust wind speeds associated with a given sustained wind speed may vary with storm size and intensity, as suggested in Vickery et al. (2009a).

Wind Speeds for Serviceability Design. For serviceability applications such as drift and habitability, the Appendix C Commentary presents maps of peak gust wind speeds at 33 ft (10 m) above ground in Exposure C conditions for return periods of 10, 25, 50, and 100 years (Figs. CC.2-1 through CC.2-4).

The probability, P_n , that the wind speed associated with a certain annual probability, P_a , will be equaled or exceeded at least once during an exposure period of n years is given by

$$P_n = 1 - (1 - P_a)^n \quad (\text{C26.5-3})$$

where

$$P_a = 1 - e^{(-1/\text{MRI})} \quad (\text{C26.5-4})$$

For MRIs of about 10 years or longer, P_a is very closely approximated by the reciprocal of the mean recurrence interval, i.e., $P_a \approx 1/\text{MRI}$.

As an example, if a wind speed is based upon $P_a = 0.02/\text{year}$ (50-year MRI), the probability that this speed will be equaled or exceeded (at least once) during a 25-year period is 0.40 (i.e., 40%), and the probability of being equaled or exceeded in a 50-year period is 64%. Similarly, if a wind speed is based upon $P_a = 0.00143$ (700-year MRI), the probability that this speed will be equaled or exceeded during a 25-year period is 3.5%, and the probability of being equaled or exceeded in a 50-year period is 6.9%.

Some products have been evaluated, and test methods have been developed based on design wind speeds that are consistent with the unfactored load effects typically used in allowable stress design (ASD). Table C26.5-7 provides a comparison of the strength design-based wind speeds used in the ASCE 7-10 and 7-16 basic wind speed maps and the ASCE 7-05 basic wind speeds used in these product evaluation reports and test methods. A column of values is also provided to allow comparison with ASCE 7-93 basic wind speeds.

C&C Rating for Building Envelope Products. Building envelope products that have been tested to air pressure standards (such as ASTM E330, CSA A123.21, or other standards that

incorporate a safety factor) are typically rated for an allowable stress design wind pressure ($0.6W$) rather than a strength design pressure ($1.0W$) or wind speed. In order to properly select products tested and rated in this manner, the C&C pressures determined from Chapter 30 should be adjusted for the allowable stress design load factor of $0.6W$ in Section 2.4.1.

C26.5.2 Special Wind Regions. Although the wind speed maps of Figs. 26.5-1 and 26.5-2 are valid for most regions of the country, there are special regions in which wind speed anomalies are known to exist. Some of these special regions are noted in Figs. 26.5-1 and 26.5-2. In 2016, the special wind regions were restored to the areas originally designated in ANSI A58.1 through ASCE 7-93. The regions had inadvertently been changed in ASCE 7-95 because of an unintended graphical error. The special wind regions around the Great Lakes in the northeast United States and in the Puget Sound area near Seattle were omitted intentionally in ASCE 7-95 because of lack of meteorological data demonstrating the existence of wind speeds higher than could be explained by exposure. Winds blowing over mountain ranges or through gorges or river valleys in these special regions can develop speeds that are substantially higher than the values indicated on the map. When selecting basic wind speeds in these special regions, use of regional climatic data and consultation with a wind engineer or meteorologist is advised.

It is also possible that anomalies in wind speeds exist on a micrometeorological scale. For example, wind speed-up over hills and escarpments is addressed in Section 26.8. Wind speeds over complex terrain may be better determined by wind tunnel studies as described in Chapter 31. Adjustments of wind speeds should be made at the micrometeorological scale on the basis of wind engineering or meteorological advice and used in accordance with the provisions of Section 26.5.3 when such adjustments are warranted. Because of the complexity of mountainous terrain and valley gorges in Hawaii, there are topographic wind speed-up effects that cannot be addressed solely by Fig. 26.8-1 (Applied Research Associates 2001). In the Hawaii special wind region, research and analysis have established that there are special K_{zt} topographic effect adjustments (Chock et al. 2005).

The southernmost special wind region in California experiences Santa Ana winds (dry, mountain downslope winds). The appropriate boundaries of this region are difficult to quantify because of a lack of data.

C26.5.3 Estimation of Basic Wind Speeds from Regional Climatic Data. When using regional climatic data in accordance with the provisions of Section 26.5.3 and in lieu of the basic wind speeds given in Figs. 26.5-1 and 26.5-2, the user is cautioned that the gust factors, velocity pressure exposure coefficients, gust-effect factors, pressure coefficients, and force coefficients of this standard are intended for use with the 3-s gust speed at 33 ft (10 m) above ground in terrain with open exposure. It is necessary, therefore, that regional climatic data based on a different averaging time, for example, hourly mean, 10-min mean, fastest minute, or fastest mile, be adjusted to reflect peak gust speeds at 33 ft (10 m) above ground in terrain with open exposure. The results of statistical studies of wind-speed records, reported by Durst (1960) for extratropical winds and for hurricanes (Vickery et al. 2000), are given in Fig. C26.5-1, which defines the relation between wind speed averaged over time t in seconds, V_t , and the hourly wind speed, V_{3600} . The hurricane simulation model described in Section C26.5.1 uses the gust-factor curve from ESDU (1982, 1993), which has been shown to be valid for hurricane winds (Vickery and Skerlj 2005). Similar conclusions regarding hurricane gust

Table C26.5-7 Basic Wind Speeds: ASCE 7-93 to ASCE 7-16

ASCE 7-95 through ASCE 7-05 Basic Wind Speed (3-s gust in mi/h)	ASCE 7-10 and ASCE 7-16 Basic Wind Speed (3-s gust, mi/h)	ASCE 7-93 and Prior Editions Basic Wind Speed (fastest mile, mi/h)
85	108 ^a	71
90	114 ^a	76
100	126	85
105	133	90
110	139	95
120	152	104
130	164	114
140	177	123
145	183	128
150	190	133
170	215	152

^aIn ASCE 7-10 wind speed values of 108 mi/h and 114 mi/h were rounded to 110 mi/h and 115 mi/h, respectively.

Note: Conversion of mi/h to m/s: $\text{mi/h} \times 0.44704 = \text{m/s}$.

factors were drawn by Jung and Masters (2013). The relation between wind speed averaging times for extratropical winds in any terrain exposure is given in Section 11.2.4.2 of Simiu (2011).

In using local data, it should be emphasized that sampling errors can lead to large uncertainties in specification of the wind speed. Sampling errors are the errors associated with the limited size of the climatological data samples (e.g., years of record of extreme speeds). It is possible to have a 20-mi/h (8.9-m/s) error in the estimated extreme wind speed at an individual station with a record length of 30 years. When short local records are used to estimate extreme wind speeds, care and conservatism should be exercised in their use.

If meteorological data are used to justify a wind speed lower than the basic wind speed from Figs. 26.5-1 and 26.5-2, an analysis of sampling error is required. This can be accomplished by showing that the difference between the estimated speed and the basic wind speed from Figs. 26.5-1 and 26.5-2 is at least two to three times the standard deviation of the sampling error (Simiu and Scanlan 1996).

C26.6 WIND DIRECTIONALITY

The wind load factor 1.3 in ASCE 7-95 included a “wind directionality factor” with a nominal value of 0.85 (Ellingwood 1981; Ellingwood et al. 1982). This factor accounts for two effects: (1) The reduced probability of maximum winds coming from any given direction, and (2) the reduced probability of the maximum pressure coefficient occurring for any given wind

direction. The nominal wind directionality factor (denoted by K_d in the standard) is tabulated in Table 26.6-1 for different structure types. As new research becomes available, this factor can be directly modified. Nominal values for the factor were established from references in the literature and collective committee judgment. A value of 0.85 might be more appropriate if a triangular trussed frame is shrouded in a round cover. A value of 0.95 might be more appropriate for a round structure that has a nonaxisymmetrical lateral load resistance system.

C26.7 EXPOSURE

The descriptions of the surface roughness categories and exposure categories in Section 26.7 have been expressed as far as possible in easily understood verbal terms that are sufficiently precise for most practical applications. Upwind surface roughness conditions required for Exposures B and D are shown schematically in Figs. C26.7-1 and C26.7-2, respectively. Aerial photographs showing examples of Exposures B, C, and D are shown in Figs. C26.7-5 through C26.7-7. For cases where the designer wishes to make a more detailed assessment of the surface roughness category and exposure category, the following more-mathematical description is offered for guidance (Irwin 2006). The ground surface roughness is best measured in terms of a roughness length parameter called z_0 . Each of the surface roughness categories B through D corresponds to a range of values of this parameter, as does the even-rougher category A used in older versions of the standard in heavily built-up urban areas but removed

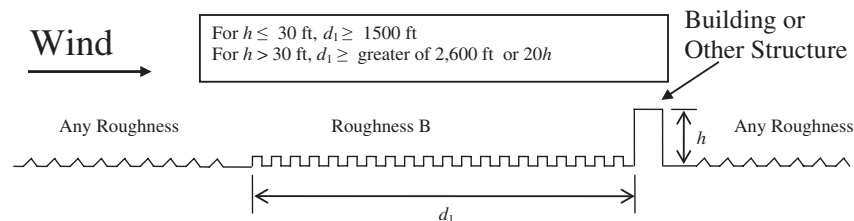
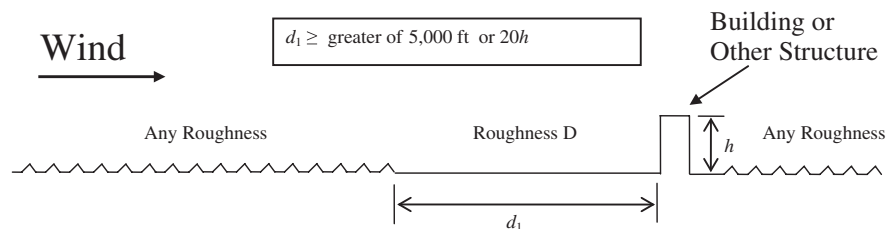
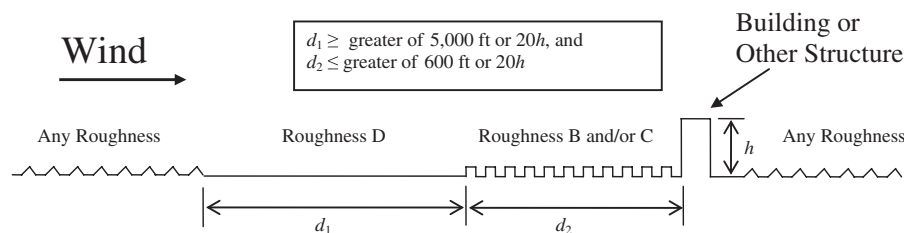


FIGURE C26.7-1 Upwind Surface Roughness Conditions Required for Exposure B



(a)



(b)

FIGURE C26.7-2 Upwind Surface Roughness Conditions Required for Exposure D, for the Cases with (a) Surface Roughness D Immediately Upwind of the Building, and (b) Surface Roughness B and/or C Immediately Upwind of the Building

Table C26.7-1 Range of z_0 by Exposure Category

Exposure Category	Lower Limit of z_0 , ft (m)	Typical Value of z_0 , ft (m)	Upper Limit of z_0 , ft (m)	z_0 Inherent in Tabulated K_z Values in Table 26.10-1, ft (m)
A	$2.3 (0.7) \leq z_0$	6.6 (2)	—	
B	$0.5 (0.15) \leq z_0$	1.0 (0.3)	$z_0 < 2.3 (0.7)$	0.66 (0.20)
C	$0.033 (0.01) \leq z_0$	0.066 (0.02)	$z_0 < 0.5 (0.15)$	0.066 (0.02)
D	—	0.016 (0.005)	$z_0 < 0.033 (0.01)$	0.016 (0.005)

in the recent editions. The range of z_0 in ft (m) for each terrain category is given in Table C26.7-1. Exposure A has been included in Table C26.7-1 as a reference that may be useful when using the Wind Tunnel Procedure. Further information on values of z_0 in different types of terrain can be found in Simiu and Scanlan (1996) and Table C26.7-2 based on Davenport et al. (2000) and Wieringa et al. (2001).

The roughness classifications in Table C26.7-2 are not intended to replace the use of exposure categories as required in the standard for structural design purposes. However, the terrain roughness classifications in Table C26.7-2 may be related

to exposure categories by comparing z_0 values between Tables C26.7-1 and C26.7-2. For example, the z_0 values for Classes 3 and 4 in Table C26.7-2 fall within the range of z_0 values for Exposure C in Table C26.7-1. Similarly, the z_0 values for Classes 5 and 6 in Table C26.7-2 fall within the range of z_0 values for Exposure B in Table C26.7-1.

Research described in Powell et al. (2003), Donelan et al. (2004), and Vickery et al. (2009a) showed that the drag coefficient over the ocean in high winds in hurricanes does not continue to increase with increasing wind speed, as was previously believed (e.g., Powell 1980). These studies showed that the sea surface drag coefficient, and hence the aerodynamic roughness of the ocean, reached a maximum at mean wind speeds of about 67 mi/h (30 m/s). There is some evidence that the drag coefficient actually decreases (i.e., the sea surface becomes aerodynamically smoother) as the wind speed increases further (Powell et al. 2003) or as the hurricane radius decreases (Vickery et al. 2009a). The consequences of these studies are that the surface roughness over the ocean in a hurricane is consistent with that of Exposure D rather than Exposure C. Consequently, the use of Exposure D along the hurricane coastline is required.

For Exposure B, the tabulated values of K_z correspond to $z_0 = 0.66$ ft (0.2 m), which is below the typical value of 1 ft (0.3 m), whereas for Exposures C and D they correspond to the typical

Table C26.7-2 Davenport Classification of Effective Terrain Roughness

Class	z_0 , ft (m) ^a	α^b	z_g , ft (m) ^b	z_g (ft or m) ^c	Wind Flow and Landscape Description ^d
1	0.0007 (0.0002)	12.9	509 (155)	$z_d = 0$	<i>Sea</i> : Open sea or lake (irrespective of wave size), tidal flat, snow-covered flat plain, featureless desert, tarmac, and concrete, with a free fetch of several kilometers.
2	0.016 (0.005)	11.4	760 (232)	$z_d = 0$	<i>Smooth</i> : Featureless land surface without any noticeable obstacles and with negligible vegetation, e.g., beaches, pack ice without large ridges, marsh, and snow-covered or fallow open country.
3	0.1 (0.03)	9.0	952 (290)	$z_d = 0$	<i>Open</i> : Level country with low vegetation (e.g., grass) and isolated obstacles with separations of at least 50 obstacle heights, e.g., grazing land without windbreaks, heather, moor, and tundra, runway area of airports. Ice with ridges across-wind.
4	0.33 (0.10)	7.7	1,107 (337)	$z_d = 0$	<i>Roughly open</i> : Cultivated or natural area with low crops or plant covers, or moderately open country with occasional obstacles (e.g., low hedges, isolated low buildings, or trees) at relative horizontal distances of at least 20 obstacle heights.
5	0.82 (0.25)	6.8	1,241 (378)	$z_d = 0.2z_H$	<i>Rough</i> : Cultivated or natural area with high crops or crops of varying height and scattered obstacles at relative distances of 12 to 15 obstacle heights for porous objects (e.g., shelterbelts) or 8 to 12 obstacle heights for low solid objects (e.g., buildings).
6	1.64 (0.5)	6.2	1,354 (413)	$z_d = 0.5z_H$	<i>Very rough</i> : Intensely cultivated landscape with many rather large obstacle groups (large farms, clumps of forest) separated by open spaces of about 8 obstacle heights. Low, densely planted major vegetation like bushland, orchards, young forest. Also, area moderately covered by low buildings with interspaces of 3 to 7 building heights and no high trees.
7	3.3 (1.0)	5.7	1,476 (450)	$z_d = 0.7z_H$	<i>Skimming</i> : Landscape regularly covered with similar-size large obstacles, with open spaces of the same order of magnitude as obstacle heights, e.g., mature regular forests, densely built-up area without much building height variation.
8	$\geq \text{matu} (\geq \text{mat})$	5.2	1,610 (490)	Analysis by wind tunnel advised	<i>Chaotic</i> : City centers with mixture of low-rise and high-rise buildings, or large forests of irregular height with many clearings. (Analysis by wind tunnel advised.)

^aThe surface roughness length, z_0 , represents the physical effect that roughness objects (obstacles to wind flow) on the Earth's surface have on the shape of the atmospheric boundary layer wind velocity profile as determined by the logarithmic law and used in the ESDU model.

^bThe power law uses α on the shape of the atmospheric boundary layer wz_g , representing the height at which geostrophic wind flow begins to occur, as the basis for determining the boundary layer wind velocity profile and velocity pressure exposure coefficients (see Section C26.10.1). The values provided in this table are based on the published z_0 values and use of Eqs. (C26.10-3) and (C26.10-4).

^cThe zero plane displacement height, z_d , is the elevation above ground that the base of the logarithmic law (and power law) wind profile must be elevated to accurately depict the boundary layer wind flow. Below z_d and less than some fraction of the typical height, z_H , of obstacles causing roughness, the near ground wind flow is characterized as a turbulent exchange with the boundary layer wind flow above resulting in significant shielding effects under uniform to moderately uniform roughness conditions (e.g., Classes 5 through 7 in this table). In this condition, the effective mean roof height, h_{eff} , may then be determined as $h - z_d$ (but not less than 15 ft or 4.6 m) for the purpose of determining MWFRS wind loads acting on a building structure located within such a roughness class. Appropriate values of z_d for a given site may vary widely, and those shown in this table should be used with professional judgment. Because of the presence of highly turbulent flow at elevations near or below z_d (except perhaps structures embedded in uniform Class 7 roughness), use of an effective mean roof height should not be applied for the determination of components and cladding wind loads. In Class 8 roughness where wind flow disruptions can be highly nonuniform, channeling effects and otherwise "chaotic" wind flow patterns can develop between and below the height of obstacles to wind flow. For this reason, a wind tunnel study is generally advised.

^dUse of these wind flow and landscape descriptions should result in no greater than one roughness class error, corresponding to a maximum $\pm$ of these wind q_h .

value of z_0 . The reason for the difference in Exposure B is that this category of terrain, which is applicable to suburban areas, often contains open patches, such as highways, parking lots, and playing fields. These open patches cause local increases in the wind speeds at their edges. By using an exposure coefficient corresponding to a lower-than-typical value of z_0 , some allowance is made for this. The alternative would be to introduce a number of exceptions to the use of Exposure B in suburban areas, which would add an undesirable level of complexity.

The value of z_0 for a particular terrain can be estimated from the typical dimensions of surface roughness elements and their spacing on the ground area using an empirical relationship, found by Lettau (1969), which is

$$z_0 = 0.5 H_{ob} \frac{S_{ob}}{A_{ob}} \quad (\text{C26.7-1})$$

where

H_{ob} = average height of the roughness in the upwind terrain;
 S_{ob} = average vertical frontal area per obstruction presented to the wind; and
 A_{ob} = average area of ground occupied by each obstruction, including the open area surrounding it.

Vertical frontal area is defined as the area of the projection of the obstruction onto a vertical plane normal to the wind direction. The area S_{ob} may be estimated by summing the approximate vertical frontal areas of all obstructions within a selected area of upwind fetch and dividing the sum by the number of obstructions in the area. The average height, H_{ob} , may be estimated in a similar way by averaging the individual heights rather than using the frontal areas. Likewise, A_{ob} may be estimated by dividing the size of the selected area of upwind fetch by the number of obstructions in it.

As an example, if the upwind fetch consists primarily of single-family homes with typical height $H_{ob} = 20$ ft (6 m), vertical frontal area (including some trees on each lot) of 1,000 ft² (100 m²), and ground area per home of 10,000 ft² (1,000 m²), then z_0 is calculated to be $z_0 = 0.5 \times 20 \times 1,000 / 10,000 = 1$ ft (0.3 m), which falls into Exposure Category B, according to Table C26.7-1.

Trees and bushes are porous and are deformed by strong winds, which reduce their effective frontal areas (ESDU 1993). For conifers and other evergreens, no more than 50% of their gross frontal area can be taken to be effective in obstructing the wind. For deciduous trees and bushes, no more than 15% of their gross frontal area can be taken to be effective in obstructing the wind. *Gross frontal area* is defined in this context as the projection onto a vertical plane (normal to the wind) of the area enclosed by the envelope of the tree or bush.

Ho (1992) estimated that the majority of buildings (perhaps as much as 60–80%) have an exposure category corresponding to Exposure B. While the relatively simple definition in the standard normally suffices for most practical applications, the designer is often in need of additional information, particularly with regard to the effect of large openings or clearings (e.g., large parking lots, freeways, or tree clearings) in the otherwise “normal” ground Surface Roughness B. The following is offered as guidance for these situations:

1. The simple definition of Exposure B given in the body of the standard, using the surface roughness category definition, is shown pictorially in Fig. C26.7-1. This definition applies for the Surface Roughness B condition prevailing 2,630 ft (800 m) upwind with insufficient “open patches,” as defined in the following procedure to disqualify the use

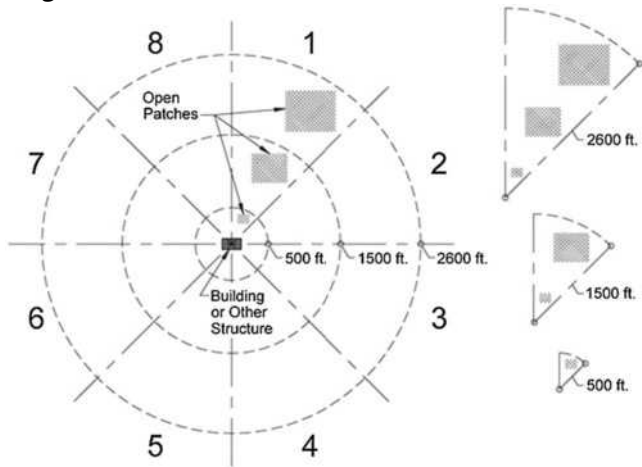
of Exposure B. This procedure on the net effect of these open patches applies where the prevailing exposure beyond 2,600 ft (792 m) is Exposure B.

2. An open area in the Surface Roughness B large enough to have a significant effect on the exposure category determination is defined as an “open patch.” To be considered an “open patch,” an open area meets the following:
 - a. Open areas should be greater than the minimum areas given by Fig. C26.7-4. Interpolation shall be used between the reference distances of 500 ft, 1,500 ft, and 2,600 ft (152, 457, and 790 m, respectively) to determine the intermediate minimum open patch area criteria.
 - b. The open area shall have minimum dimensions given by conditions i, ii, or iii below and have length to width ratios between 0.5 and 2.0.
 - i. Within 500 ft (152 m) of the building or structure, an open area greater than or equal to approximately 164 ft (50 m) in length or width.
 - ii. At 1,500 ft (457 m) upwind from the building or structure, an open area greater than or equal to approximately 328 ft (100 m) in length or width.
 - iii. At 2,600 ft (790 m) upwind from the building or structure, an open area greater than or equal to approximately 500 ft (152 m) in length or width.
3. Open patches separated by less than the along-wind dimension of the larger patch shall be treated as equivalent to a single open patch with length equal to the sum of the individual patch along-wind dimensions and width determined to provide an area equal to the sum of the individual open patch areas.
4. A circular sector is an area defined by a limiting radius from the center and an arc, in this case, 45° per Section 26.7.4. If the proportion of open patch within any of the sectors defined by the three radii above is less than 25% of the sector area, the sector is considered to meet the requirements for Exposure Category B. Where the proportion of open patches within any 45° sector within any of the three radii of 500 ft (152 m), 1,500 ft (457 m), or the greater of 2,600 ft (790 m) or 20 times the height of the building or structure exceeds 25% of the sector area but is not greater than 50%, the values of K_z are taken as the average of the Exposure B and C values within 100 ft (31 m) height above grade. Above 100 ft (31 m), Exposure B values of K_z shall still apply. Where the proportion of open patches within any of the sectors defined by the three radii of the building or structure exceeds 50%, the values of K_z shall be based on Exposure C.
5. The procedure for evaluation of the net effect of open patches of Surface Roughness C or D on the use of Exposure Category B is shown pictorially in Figs. C26.7-3 and C26.7-4. Note that the plan location of any open patch may have a different effect for different wind directions.

This above procedure is a simplification derived from a boundary-layer model, and therefore more exact results for the velocity profile may be achieved through direct use of an accepted boundary-layer model that is capable of addressing the effects of open areas within a regime defined by the surface roughness parameters given in Table C26.7-2.

Aerial photographs, representative of each exposure type, are included in Figs. C26.7-5 to C26.7-7 to aid the user in establishing the proper exposure for a given site. Obviously, the proper assessment of the exposure is a matter of good engineering judgment. This fact is particularly true in light of the possibility that the exposure could change in one or more wind directions due to future demolition and/or development.

Diagrams

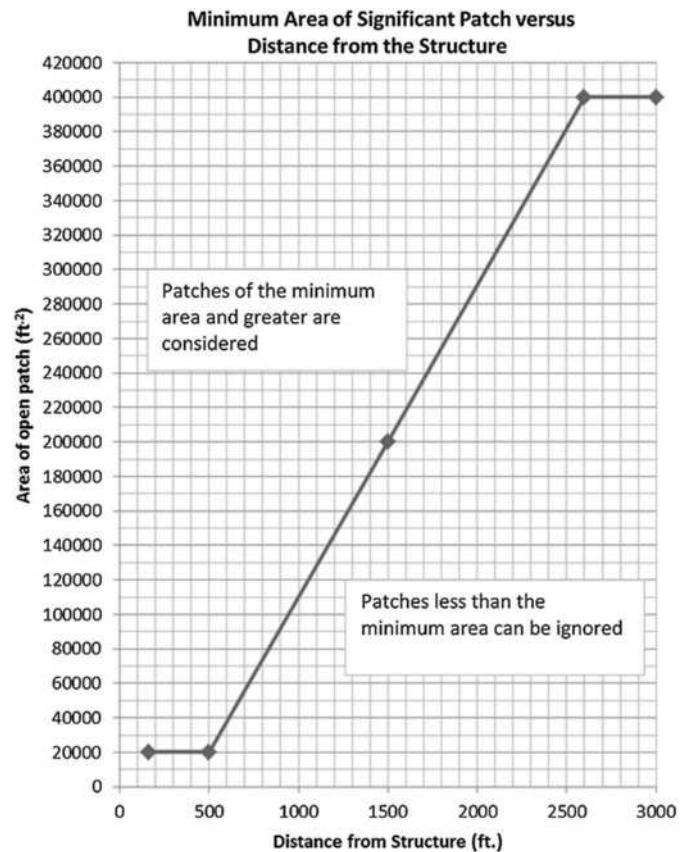


Notes

1. For each selected wind direction at which the wind loads are to be determined, the exposure of the building or structure shall be determined for the two upwind sectors extending 45° either side of the selected wind direction.
2. Consider open patches of sizes equal to or greater than the areas given in Figure C26.7-4 per Commentary Section C26.7.
3. Determine the proportion of open patches in any 45° sector within radii of 500 ft, 1,500 ft, or the greater of 2,600 ft or 20 times the height of the structure.
4. If the proportion of open patch within any of the three radii above is less than 25% of the sector area, the sector is considered to meet the requirements for Exposure B. Where the proportion within any of the three radii above exceeds 25% of the sector area but is not greater than 50%, the values of K_z are taken as the average of the Exposure B and C values within 100 ft height above grade. Above 100 ft, Exposure B values shall still apply. Where the proportion of open patches within any of the three radii of the structure exceeds 50%, the values of K_z shall be based on Exposure C.
5. Apply the exposure requirements of Section 26.7.4 once the directional exposures are determined for each sector. See Commentary Section C26.7.4.

**FIGURE C26.7-3 Exposure B with Upwind Open Patches
Sector Analysis**

C26.7.4 Exposure Requirements. Section 26.5.1 of the standard requires that a structure be designed for winds from all directions. A rational procedure to determine directional wind loads is described here. Wind load for buildings using Section 27.3.1 and Figs. 27.3-1, 27.3-2, or 27.3-3 are determined for eight wind directions at 45° intervals, with four falling along primary building axes, as shown in Fig. C26.7-8. For each of the eight directions, upwind exposure is determined for each of two 45° sectors, one on each side of the wind direction axis. The sector with the exposure giving highest loads is used to define wind loads for that direction. For example, for winds from the north, the exposure from Sector 1 or 8, whichever gives the higher load, is used. For wind from the east, the exposure from Sector 2 or 3, whichever gives the highest load, is used. For wind coming from the northeast, the more exposed of Sectors 1 or 2 is used to determine full x and y loading individually, and then 75% of these loads are to be applied in each direction at the same time, according to the requirements of Section 27.3.5 and Fig. 27.3-8. The procedure defined in this section for determining wind loads in each design direction is not to be confused with the determination of the wind directionality factor, K_d . The K_d



**FIGURE C26.7-4 Minimum Area of Individual Open Patches
Affecting Qualification of Exposure B**

factor determined from Section 26.6 and Table 26.6-1 applies for all design wind directions. See Section C26.6.

C&C loads for all buildings and MWFRS loads for low-rise buildings are determined using the upwind exposure for the single surface roughness in one of the eight sectors of Fig. C26.7-8 that gives the highest wind loads.

C26.8 TOPOGRAPHIC EFFECTS

This section specifies when topographic effects need to be applied to a particular structure (Means et al. 1996). In an effort to exclude situations where little or no topographic effect exists, Condition 2 recognizes the fact that the topographic feature should protrude significantly above (by a factor of 2 or more) upwind terrain features before it becomes a factor. For example, if a significant upwind terrain feature has a height of 35 ft (10.7 m) above its base elevation and has a top elevation of 100 ft (30.5 m) above mean sea level, then the topographic feature (hill, ridge, or escarpment) must have at least the H specified and extend to elevation 170 ft (52 m) above mean sea level [$100 \text{ ft} + (2 \times 35 \text{ ft})$] or [$31 \text{ m} + (2 \times 10 \text{ m})$] in S.I. within the 2-mi (3.2-km) radius specified.

Condition 5 specifies a lower height, H , for consideration of topographic effects in Exposure C and D than for Exposure B (Means et al. 1996), and observation of actual wind damage has shown that the affected height, H , is less in Exposures C and D.

Buildings sited on the upper half of an isolated hill or escarpment may experience significantly higher wind speeds than buildings situated on level ground. The topographic feature (2D ridge or escarpment, or 3D axisymmetrical hill) is described



FIGURE C26.7-5(a) Exposure B: Suburban Residential Area with Mostly Single-Family Dwellings. Low-Rise Structures, Less Than 30 ft (9.1 m) High, in the Center of the Photograph Have Sites Designated as Exposure B with Surface Roughness Category B Terrain around the Site for a Distance Greater Than 1,500 ft (457 m) in Any Wind Direction



FIGURE C26.7-5(b) Exposure B: Urban Area with Numerous Closely Spaced Obstructions Having the Size of Single-Family Dwellings or Larger. For All Structures Shown, Terrain Representative of Surface Roughness Category B Extends More Than 20 Times the Height of the Structure or 2,600 ft (792 m), Whichever Is Greater, in the Upwind Direction

by two parameters, H and L_h . H is the height of the hill or difference in elevation between the crest and that of the upwind terrain. L_h is the distance upwind of the crest to where the ground elevation is equal to half the height of the hill. K_z is determined from three multipliers, K_1 , K_2 , and K_3 , which are obtained from

Fig. 26.8-1. K_1 is related to the shape of the topographic feature; the maximum speed-up near the crest, K_2 , accounts for the reduction in speed-up with distance upwind or downwind of the crest; and K_3 accounts for the reduction in speed-up with height above the local ground surface.



FIGURE C26.7-5(c) Exposure B: Structures in the Foreground Are Located in Exposure B. Structures in the Center Top of the Photograph Adjacent to the Clearing to the Left, Which Is Greater Than Approximately 656 ft (200 m) in Length, are Located in Exposure C When Wind Comes from the Left over the Clearing. (See Figure C26.7-4)



FIGURE C26.7-6(a) Exposure C: Flat Open Grassland with Scattered Obstructions Having Heights Generally Less Than 30 ft (9.1 m)

The multipliers listed in Fig. 26.8-1 are based on the assumption that the wind approaches the hill along the direction of maximum slope, causing the greatest speed-up near the crest. The average maximum upwind slope of the hill is approximately

$H/2L_h$, and measurements have shown that hills with slopes of less than about 0.10 ($H/L_h < 0.20$) are unlikely to produce significant speed-up of the wind. For values of $H/L_h > 0.5$, the speed-up effect is assumed to be independent of slope. The



FIGURE C26.7-6(b) Exposure C: Open Terrain with Scattered Obstructions Having Heights Generally Less Than 30 ft (9.1 m). For Most Wind Directions, all One-Story Structures with a Mean Roof Height Less Than 30 ft (9.1 m) in the Photograph are Less Than 1,500 ft (457 m) or 10 Times the Height of the Structure, Whichever Is Greater, from an Open Field that Prevents the Use of Exposure B



FIGURE C26.7-7 Exposure D: A Building at the Shoreline (Excluding Shorelines in Hurricane-Prone Regions) with wind Flowing over Open Water for a Distance of at Least One Mile. Shorelines in Exposure D Include Inland Waterways, the Great Lakes, and Coastal Areas of California, Oregon, Washington, and Alaska

speed-up principally affects the mean wind speed rather than the amplitude of the turbulent fluctuations, and this fact has been accounted for in the values of K_1 , K_2 , and K_3 given in Fig. 26.8-1. Therefore, values of K_z obtained from Fig. 26.8-1 are intended

for use with velocity pressure exposure coefficients, K_h and K_z , which are based on gust speeds.

It is not the intent of Section 26.8 to address the general case of wind flow over hilly or complex terrain for which engineering

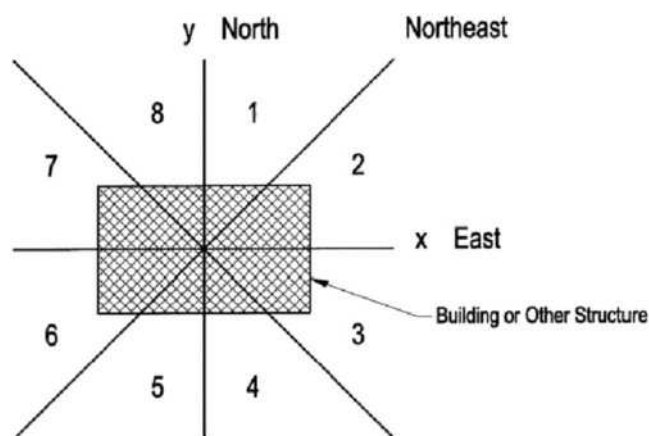


FIGURE C26.7-8 Determination of Wind Loads from Different Directions

judgment, expert advice, or the Wind Tunnel Procedure as described in Chapter 31 may be required. Background material on topographic speed-up effects may be found in the literature (Jackson and Hunt 1975; Lemelin et al. 1988; Walmsley et al. 1986).

The designer is cautioned that, at present, the standard contains no provision for vertical wind speed-up because of a topographic effect, even though this phenomenon is known to exist and can cause additional uplift on roofs. Additional research is required to quantify this effect before it can be incorporated into the standard.

C26.9 GROUND ELEVATION FACTOR

The ratio of air pressure and density at elevation z relative to the standard values at $z=0$, with constant temperature, is given by the barometric formula

$$p_z/p_0 = \rho_z/\rho_0 = e^{-gz/RT}$$

where

g = acceleration of gravity = 32.174 ft/s² (9.807 m/s²),
 R = gas constant of air = 1,718 lb-ft/slug/°R (287 N-m/kg/K),
 and
 T = absolute temperature = 518°R (288K).

With these values and elevation z = elevation of ground above sea level, the ratio is determined from the formulas given in Table 26.9-1 where $K_e = p_z/\rho_0$. For reference, a more complete

Table C26.9-1. Gound Elevation Factor including Air Density

Ground elevation, z_g		Air density, ρ		Ratio, K_e
ft	m	slug/ft ³	kg/m ³	ρ/ρ_0
-10,000	-305	0.000247	1.269	1.04
0	0	0.000238	1.224	1.00
1,000	305	0.000229	1.180	0.96
2,000	610	0.000221	1.138	0.93
3,000	914	0.000213	1.098	0.90
4,000	1,219	0.000206	1.059	0.86
5,000	1,524	0.000198	1.021	0.83
6,000	1,829	0.000191	0.985	0.80
7,000	2,134	0.000185	0.950	0.78
8,000	2,438	0.000178	0.916	0.75
9,000	2,743	0.000172	0.883	0.72
10,000	3,048	0.000166	0.852	0.70

version of Table 26.9-1 including air density values is provided in Table C26.9-1.

$K_e = 1.0$ is permitted in all cases. While this is somewhat unconservative for elevations below sea level, the committee believes it is reasonable to permit this since the effect is very small for all areas below sea level in the U.S. (0 to -300 ft in Death Valley, a maximum of 1 percent increase in air density), and is likely to be reduced even more due to higher average temperatures.

C26.10 VELOCITY PRESSURE

C26.10.1 Velocity Pressure Exposure Coefficient. The velocity pressure exposure coefficient K_z can be obtained using the equations:

$$K_z = 2.01 \left(\frac{z}{z_g} \right)^{2/\alpha} \quad \text{for } 15 \text{ ft} \leq z \leq z_g \quad (\text{C26.10-1})$$

$$K_z = 2.01 \left(\frac{z}{z_g} \right)^{2/\alpha} \quad \text{for } 4.6 \text{ m} \leq z \leq z_g \quad (\text{C26.10-1.si})$$

$$K_z = 2.01 \left(\frac{15}{z_g} \right)^{2/\alpha} \quad \text{for } z \leq 15 \text{ ft} \quad (\text{C26.10-2})$$

$$K_{zu} = 2.01 \left(\frac{20}{z_g} \right)^{2/\alpha} \quad \text{for } z \leq 4.6 \text{ m} \quad (\text{C26.10-2.si})$$

in which values of α and z_g are given in Table 26.11-1. These equations are now given in Table 26.10-1 to aid the user.

The values of α given in Table 26.11-1 define gust profiles. The mean profiles implied by these α values are based on 4.5, 7.0, and 10 for Exposures B, C, and D, respectively. These have been in use for the underlying mean velocity provided for K_z since 1982. The α values in Table 26.11-1 based on 4.0, 6.5, and 9.0 are used only to calculate the gust factor, G_f , for flexible structures in Section 26.11.5.

Changes were implemented in ASCE 7-98, including truncation of K_z values for Exposures A and B below heights of 100 ft (30.5 m) and 30 ft (9.1 m), respectively, applicable to Components and Cladding (C&C) and the Envelope Procedure. Exposure A was eliminated in the 2002 edition.

In the ASCE 7-05 standard, the K_z expressions were unchanged from ASCE 7-98. However, the possibility of interpolating between the standard exposures using a rational method was added in the ASCE 7-05 edition. One rational method is provided in the following text.

To a reasonable approximation, the empirical exponent α and gradient height z_g in the preceding expressions [Eqs. (C26.10-1) and (C26.10-2)] for exposure coefficient K_z may be related to the roughness length z_0 (where z_0 is defined in Section C26.7) by the relations

$$\alpha = c_1 z_0^{-0.133} \quad (\text{C26.10-3})$$

and

$$z_g = c_2 z_0^{0.125} \quad (\text{C26.10-4})$$

where

$c_1 = 5.65$ and $c_2 = 450$, when units of z_0 and z_g are m, and
 $c_1 = 6.62$ and $c_2 = 1,273$, when units of z_0 and z_g are ft.

The preceding relationships are based on matching the ESDU boundary-layer model (ESDU 1982, 1993) empirically with the power law relationship in Eqs. (C26.10-1) and (C26.10-2), the

ESDU model being applied at latitude 35° with a gradient wind of 168 mi/h (75 m/s). If z_0 has been determined for a particular upwind fetch, Eqs. (C26.10-1) through (C26.10-4) can be used to evaluate K_z . The correspondence between z_0 and the parameters α and z_g implied by these relationships does not align exactly with that described in the commentary to ASCE 7-95 and 7-98. However, the differences are relatively small and not of practical consequence. The ESDU boundary-layer model has also been used to derive the following simplified method (Irwin 2006) of evaluating K_z following a transition from one surface roughness to another. For more precise estimates, the reader is referred to the original ESDU model (Harris and Deaves 1981; ESDU 1990, 1993).

In uniform terrain, the wind travels a sufficient distance over the terrain for the planetary boundary layer to reach an equilibrium state. The exposure coefficient values in Table 26.11-1 are intended for this condition. Suppose that the site is a distance x miles downwind of a change in terrain. The equilibrium value of the exposure coefficient at height z for the terrain roughness downwind of the change will be denoted by K_{zd} , and the equilibrium value for the terrain roughness upwind of the change will be denoted by K_{zu} . The effect of the change in terrain roughness on the exposure coefficient at the site can be represented by adjusting K_{zd} by an increment ΔK , thus arriving at a corrected value K_z for the site:

$$K_z = K_{zd} + \Delta K \quad (\text{C26.10-5})$$

In this expression, ΔK is calculated using

$$\Delta K = (K_{33,u} - K_{33,d}) \frac{K_{zd}}{K_{33,d}} F_{\Delta K}(x)$$

$$|\Delta K| \leq |K_{zu} - K_{zd}| \quad (\text{C26.10-6})$$

where $K_{33,d}$ and $K_{33,u}$ are, respectively, the downwind and upwind equilibrium values of exposure coefficient at 33 ft (10 m) height, and the function $F_{\Delta K}(x)$ is given by

$$F_{\Delta K}(x) = \frac{\log_{10}(\frac{x_1}{x})}{\log_{10}(\frac{x_1}{x_0})} \quad (\text{C26.10-7})$$

For $x_0 < x < x_1$:

$$F_{\Delta K}(x) = 1 \quad \text{for } x < x_0$$

$$F_{\Delta K}(x) = 0 \quad \text{for } x > x_1$$

In the preceding relationships,

$$x_0 = c_3 \times 10^{-(K_{33,d} - K_{33,u})^2 - 2.3} \quad (\text{C26.10-8})$$

The constant $c_3 = 0.621$ mi (1.0 km). The length $x_1 = 6.21$ mi (10 km) for $K_{33,d} < K_{33,u}$ (wind going from smoother terrain upwind to rougher terrain downwind) or $x_1 = 62.1$ mi (100 km) for $K_{33,d} > K_{33,u}$ (wind going from rougher terrain upwind to smoother terrain downwind).

The above description is in terms of a single roughness change. The method can be extended to multiple roughness changes. The extension of the method is best described by an example. Fig. C26.10-1 shows wind with an initial profile characteristic of Exposure D encountering an expanse of B roughness, followed by a further expanse of D roughness and then some more B roughness again before it arrives at the building site. This situation is representative of wind from the sea flowing over an outer strip of land, then a coastal waterway, and then some suburban roughness before arriving at the building site. The above method for a single roughness change is first used to

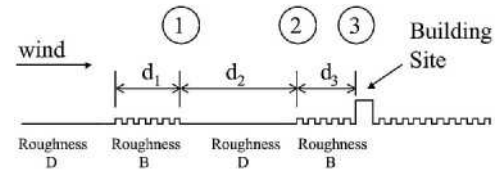


FIGURE C26.10-1 Multiple Roughness Changes Due to Coastal Waterway

compute the profile of K_z at Station 1 in Fig. C26.10-1. Call this profile $K_z^{(1)}$. The value of ΔK for the transition between Stations 1 and 2 is then determined using the equilibrium value of $K_{33,u}$ for the roughness immediately upwind of Station 1, i.e., as though the roughness upwind of Station 1 extended to infinity. This value of ΔK is then added to the equilibrium value $K_{zd}^{(2)}$ of the exposure coefficient for the roughness between Stations 1 and 2 to obtain the profile of K_z at Station 2, which we will call $K_z^{(2)}$. Note however, that the value of $K_z^{(2)}$ in this way cannot be any lower than $K_z^{(1)}$. The process is then repeated for the transition between Stations 2 and 3. Thus, ΔK for the transition from Station 2 to Station 3 is calculated using the value of $K_{33,u}$ for the equilibrium profile of the roughness immediately upwind of Station 2, and the value of $K_{33,d}$ for the equilibrium profile of the roughness downwind of Station 2. This value of ΔK is then added to $K_{zd}^{(2)}$ to obtain the profile $K_z^{(3)}$ at Station 3, with the limitation that the value of $K_z^{(3)}$ cannot be any higher than $K_z^{(2)}$.

Example 1: Single Roughness Change. Suppose that the building is 66 ft high, and its local surroundings are suburban with a roughness length $z_0 = 1$ ft. However, the site is 0.37 mi downwind of the edge of the suburbs, beyond which the open terrain is characteristic of open country with $z_0 = 0.066$ ft. From Eqs. (C26.10-1), (C26.10-3), and (C26.10-4), for the open terrain,

$$\alpha = c_1 z_0^{-0.133} = 6.62 \times 0.066^{-0.133} = 9.5$$

$$z_g = c_2 z_0^{0.125} = 1,273 \times 0.066^{0.125} = 906 \text{ ft}$$

Therefore, applying Eq. (C26.10-1) at 66 ft (20 m) and 33 ft (10 m) heights,

$$K_{zu} = 2.01 \left(\frac{66}{906} \right)^{2/9.5} = 1.16$$

and

$$K_{33,u} = 2.01 \left(\frac{33}{906} \right)^{2/9.5} = 1.00$$

Similarly, for the suburban terrain,

$$\alpha = c_1 z_0^{-0.133} = 6.62 \times 1.0^{-0.133} = 6.62$$

$$z_g = c_2 z_0^{0.125} = 1,273 \times 1.0^{0.125} = 1,273 \text{ ft}$$

Therefore,

$$K_{zd} = 2.01 \left(\frac{66}{1,273} \right)^{2/6.62} = 0.82$$

and

$$K_{33,d} = 2.01 \left(\frac{33}{1,273} \right)^{2/6.62} = 0.67$$

From Eq. (C26.10-8),

$$x_0 = c_3 \times 10^{-(K_{33,d} - K_{33,u})^2 - 2.3} = 0.621 \\ \times 10^{-(0.62 - 1.0)^2 - 2.3} = 0.00241 \text{ mi}$$

From Eq. (C26.10-7),

$$F_{\Delta K}(x) = \frac{\log_{10}(6.21/0.36)}{\log_{10}(6.21/0.00241)} = 0.36$$

Therefore, from Eq. (C26.10-6),

$$\Delta K = (1.00 - 0.67) \frac{0.82}{0.67} 0.36 = 0.15$$

Note that because $|\Delta K|$ is 0.15, which is less than the 0.38 value of $\Delta|K_{33,u} - K_{33,d}|$, 0.15 is retained. Finally, from Eq. (C26.10-5), the value of K_z is

$$K_z = K_{zd} + \Delta K = 0.82 + 0.15 = 0.97$$

Because the value 0.97 for K_z lies between the values 0.88 and 1.16, which would be derived from Table 26.11-1 for Exposures B and C, respectively, it is an acceptable interpolation. If it falls below the Exposure B value, then the Exposure B value of K_z is to be used. The value $K_z = 0.97$ may be compared with the value 1.16 that would be required by the simple 2,600-ft fetch length requirement of Section 26.7.3.

The most common case of a single roughness change where an interpolated value of K_z is needed is for the transition from Exposure C to Exposure B, as in the example just described. For this particular transition, using the typical values of z_0 of 0.066 ft and 1.0 ft, the preceding formulas can be simplified to

$$K_z = K_{zd} \left(1 + 0.146 \log_{10} \left(\frac{6.21}{x} \right) \right) \\ K_{zB} \leq K_z \leq K_{zC} \quad (\text{C26.10-9})$$

where x is in miles, and K_{zd} is computed using $\alpha = 6.62$. K_{zB} and K_{zC} are the exposure coefficients in the standard Exposures B and C, respectively. Fig. C26.10-2 illustrates the transition from terrain roughness C to terrain roughness B from this expression. Note that it is acceptable to use the typical z_0 rather than the lower limit for Exposure B in deriving this

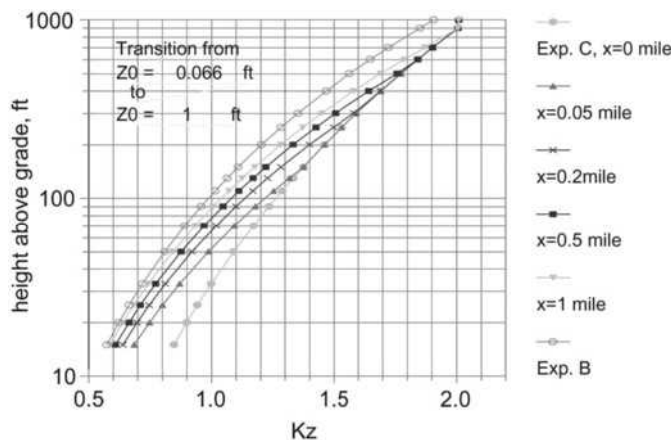


FIGURE C26.10-2 Transition from Terrain Roughness C to Terrain Roughness B, Eq. (26.10-9)

Table C26.10-1 Tabulated Exposure Coefficients

Transition from sea to	$K_{33,u}$	$K_{33,d}$	$K_{50,d}$	$F_{\Delta K}$	ΔK_{50}	$K_{50}^{(1)}$
Station 1	1.215	0.667	0.758	0.220	0.137	0.895
Transition from Station 1 to	$K_{33,u}$	$K_{33,d}$	$K_{50,d}$	$F_{\Delta K}$	ΔK_{50}	$K_{50}^{(2)}$
Station 2	0.667	1.215	1.301	0.324	-0.190	1.111
Transition from Station 2 to	$K_{33,u}$	$K_{33,d}$	$K_{50,d}$	$F_{\Delta K}$	ΔK_{50}	$K_{50}^{(3)}$
Station 3	1.215	0.667	0.758	0.498	0.310	1.067

Note: The equilibrium values of the exposure coefficients, $K_{33,u}$, $K_{33,d}$ and $K_{50,d}$ (downwind value of K_z at 50 ft), were calculated from Eq. (C26.10-1) using α and z_g values obtained from Eqs. (C26.10-3) and (C26.10-4) with the roughness values given. Then $F_{\Delta K}$ is calculated using Eqs. (C26.10-7) and (C26.10-8), and then the value of ΔK at 50-ft height, ΔK_{50} , is calculated from Eq. (C26.10-6). Finally, the exposure coefficient at 50 ft at Station i , $K_{50}^{(i)}$, is obtained from Eq. (C26.10-5).

formula because the rate of transition of the wind profiles is dependent on average roughness over significant distances, not local roughness anomalies. The potential effects of local roughness anomalies, such as parking lots and playing fields, are covered by using the standard Exposure B value of exposure coefficient, K_{zB} , as a lower limit to the calculated value of K_z .

Example 2: Multiple Roughness Change. Suppose we have a coastal waterway situation, as illustrated in Fig. C26.10-1, where the wind comes from open sea with roughness type D, for which we assume $z_0 = 0.01$ ft and passes over a strip of land 1 mi wide, which is covered in buildings that produce typical B type roughness, i.e., $z_0 = 1$ ft. It then passes over a 2-mi-wide strip of coastal waterway where the roughness is again characterized by the open water value $z_0 = 0.01$ ft. It then travels over 0.1 mi of roughness type B ($z_0 = 1$ ft) before arriving at the site, Station 3 in Fig. C26.10-1, where the exposure coefficient is required at the 50-ft height. The exposure coefficient at Station 3 at a 50-ft height is calculated as shown in Table C26.10-1.

The value of the exposure coefficient at 50 ft at Station 3 is seen from the table to be 1.067. This is above that for Exposure B, which would be 0.81, but well below that for Exposure D, which would be 1.27, and similar to that for Exposure C, which would be 1.09.

C26.10.2 Velocity Pressure. The basic wind speed is converted to velocity pressure q_z in lb/ft² (N/m²) at height z by the use of Eq. (26.10-1).

The constant in this equation reflects the mass density of air for the standard atmosphere, that is, temperature of 59°F (15°C) and sea level pressure of 29.92 in. of mercury (101.325 kPa), equal to 0.0765 lbm/ft³ or 0.002378 slug/ft³ or 0.002378 lb-s²/ft⁴ (1.225 kg/m³), and dimensions associated with wind speed in mi/h (m/s). The constant is obtained as follows:

Dynamic pressure from Bernoulli's law:

$$p = \frac{1}{2} \rho V^2$$

with V in mi/h:

$$p = \frac{1}{2} (0.002378 \text{ lb s}^2/\text{ft}^4) [V \text{ mi/h} (88 \text{ ft/s} / 60 \text{ mi/h})^2] \\ = 0.00256 V^2 \text{ lb/ft}^2$$

with V in m/s:

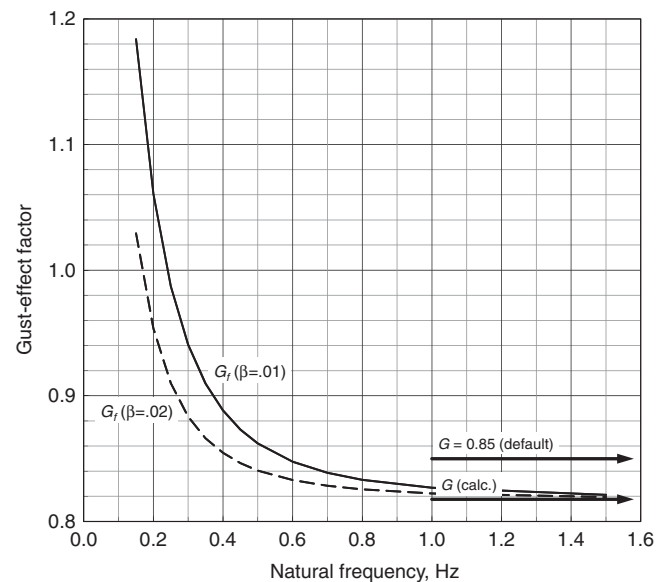
$$p = \frac{1}{2} (1.225 \text{ kg/m}^3) (1 \text{ N/1 kg m/s}^2) (V \text{ m/s})^2 \\ = 0.613 V^2 \text{ N/m}^2 \quad (\text{si})$$

Values of air density other than the standard-atmosphere values above may be adjusted using the factor K_e as described in Section C26.9.

Table C26.11-1. Gust-Effect Factor, Example

Example Calculation—Gust-Effect Factors			Example Calculation—Gust-Effect Factors (Continued)		
Item	Value	Source	Item	Value	Source
DEFAULT FACTOR FOR RIGID BUILDING (Requires $n_1 \geq 1$)			η_h Vertical decay parameter	4.002	26.9.5: $4.6n_1h/\bar{V}_z$
G Gust-effect factor	0.85	26.9-1	η_B Cross-wind decay parameter	0.667	26.9.5: $4.6n_1B/\bar{V}_z$
ALTERNATE CALCULATED FACTOR FOR RIGID BUILDING (requires $n_1 \geq 1$)			η_L Along-wind decay parameter	2.233	26.9.5: $15.4n_1L/\bar{V}_z$
h Mean roof height	600 ft (183 m)	User spec	R_h Resonant factor for h	0.219	(26.9-15a)
B Width normal to wind	100 ft (30 m)	User spec	R_B Resonant factor for B	0.671	(26.9-15a)
D Depth parallel to wind	100 ft (30 m)	User spec	R_L Resonant factor for L	0.349	(26.9-15a)
$\bar{z}$ Effective structure height	360 ft (110 m)	0.6 h (26.9-4)	R^2 Resonant response (squared)	1.313	(26.9-12)
Exposure category	B		g_R Resonant peak factor	3.787	(26.9-11)
c Turbulence intensity at 10 m	0.3	Table 26.9-1	G_f Gust-effect factor	1.173	(26.9-10)
I_z Turbulence intensity at eff. height	0.201	(26.9-7)			
I Turbulence length scale at 10 m	320 ft (98 m)	Table 26.9-1			
$\bar{e}$ Power law exponent of turbulent length scale profile	1/3	Table 26.9-1			
L_z Turbulence length scale at eff. height	710 ft (216 m)	(26.9-9)			
Q^2 Background response (squared)	0.616	(26.9-8)			
g_Q Background load peak factor	3.4	26.9-4			
g_v Velocity peak factor	3.4	26.9-4			
G Calculated gust-effect factor	0.818	(26.9-6)			
ADDITIONAL CALCULATIONS FOR FLEXIBLE BUILDING (all n_1)					
V Basic wind speed	115 mph (51 m/s)				
n_1 Fundamental natural frequency in direction of wind	0.2 Hz	Analysis or rational approximation			
β Damping ratio	0.01	Rational assignment			
$\bar{\alpha}$ Power law exponent of mean wind speed profile	0.25	Table 26.9-1			
$\bar{b}$ Gust factor 1/ F at 10 m	0.45	Table 26.9-1			
$\bar{V}_z$ Mean wind speed at effective height	138 ft/s (42.1 m/s)	26.9-16			
N_1 Reduced natural frequency	1.029	26.9-14			
R_n Resonance response factor for n	0.129	26.9-13			

Gust-effect factor relations for example building in extended frequency range:



C26.11 GUST EFFECTS

This standard specifies a single, conservative, gust-effect factor of 0.85 for rigid buildings. As an option, the designer can incorporate specific features of the wind environment and building size to more accurately calculate an alternate but more accurate gust-effect factor that accounts for the decorrelation of wind gusts over the size of the structure. One such procedure is located in the body of the standard (Solari and Kareem 1998). Neither of these factors accounts for dynamic amplification caused by vibration of the structure, but they are considered acceptable for rigid structures as defined in the standard. The alternate calculated gust factor is 5–10% lower than the value of 0.85 permitted in the standard without calculation.

A third gust-effect factor, G_f , is provided for flexible buildings and structures that do not meet the requirements of rigid structures that the fundamental natural frequency, n_1 , is greater than or equal to 1 Hz. This factor also accounts for the building size and gust size in the same manner as the alternate calculated factor for rigid buildings, but it also accounts for dynamic amplification caused by the design wind speed, the fundamental natural frequency of vibration, and the damping ratio.

Example: Calculation of the gust-effect factors for a subject building is demonstrated in Table C26.11-1. The frequency-dependent relationship among all factors is illustrated in the graph at the end of this table. The flexible factor, G_f , may be used for all cases but is required when $n_1 < 1$. This factor gradually approaches the alternate calculated factor for rigid cases, G , as the natural frequency exceeds 1, especially for higher levels of damping, but it always exceeds G . The difference is deemed negligible for n_1 greater than 1, so G , which is considerably simpler to calculate, is offered as an acceptable alternative. The default value of $G=0.85$, which requires no calculation, is offered as an even more convenient alternative when $n_1 > 1$, if the greater conservatism is acceptable to the designer. In addition, the default value results in a large abrupt change in the gust-effect value for cases that have a natural frequency close to 1, which may be awkward for a designer to reconcile. A designer is free to use any other rational procedure in the approved literature, as stated in Section 26.11.5.

The gust-effect factors account for loading effects in the along-wind direction caused by wind turbulence–structure interaction. They do not include allowances for across-wind loading effects, vortex shedding, instability caused by galloping or flutter, or amplification of aerodynamic torsion caused by building

vibration in a pure torsional mode. For structures susceptible to loading effects that are not accounted for in the gust-effect factor, information should be obtained from recognized literature (Kareem 1992, 1985; Gurley and Kareem 1993; Solari 1993a, b; Zhou et al. 2002; Chen and Kareem 2004; Bernardini et al. 2013a) or from wind tunnel tests.

Along-Wind Response. The maximum along-wind displacement response can be approximated by a static analysis of the structure under the action of loads multiplied by the appropriate gust-effect factor, as defined in the standard. Such displacements are based on the static elastic curve of the structure and are reasonably accurate when the resonant response is small compared to the mean and background responses. For highly flexible structures, where the response is dominated by resonance, more accurate values, including variation with height and dynamic responses such as acceleration, can be calculated as described in the following sections. These response components are needed for strength and serviceability limit states.

The maximum along-wind displacement as a function of height above the ground surface is given by

$$X_{\max}(z) = \frac{\phi(z)\rho B h C_{fx} \hat{V}_z^2}{2m_1(2\pi n_1)^2} K G_f \quad (\text{C26.11-1})$$

where

$\phi(z)$ = fundamental model shape, $\phi(z) = (z/h)^\xi$;

ξ = mode shape power-law exponent;

ρ = air density;

C_{fx} = mean along-wind force coefficient;

m_1 modal mass = $\int_0^h \mu(z) \phi^2(z) dz$;

n_1 = fundamental natural frequency;

$\mu(z)$ = mass per unit height: $K = (1.65)^\alpha / (\alpha + \xi + 1)$; and

$\hat{V}_z$ = 3-s gust speed at height $\bar{z}$.

$$\hat{V}_z = \hat{b}(z/33)^\alpha V \left(\frac{88}{60} \right)$$

where V is the 3-s gust speed in Exposure C (mi/h) at the reference height (obtained from Figs. 26.5-1 and 26.5-2); $\hat{b}$ and α are given in Table 26.11-1.

The root-mean-square (rms) along-wind acceleration $\sigma_{\ddot{x}}(z)$ as a function of height above the ground surface is given by

$$\sigma_{\ddot{x}}(z) = \frac{0.85 \phi(z) \rho B h C_{fx} \bar{V}_z^2}{m_1} I_z K R \quad (\text{C26.11-2})$$

where $\bar{V}_z$ is the mean hourly wind speed at height $\bar{z}$, ft/s.

$$\bar{V}_z = \bar{b} \left(\frac{\bar{z}}{33} \right)^{\bar{\alpha}} V \left(\frac{88}{60} \right) \quad (\text{C26.11-3})$$

where $\bar{b}$ and $\bar{\alpha}$ are defined in Table 26.11-1.

The maximum along-wind acceleration as a function of height above the ground surface is given by

$$\ddot{X}_{\max}(z) = g_{\ddot{x}} \sigma_{\ddot{x}}(z) \quad (\text{C26.11-4})$$

$$g_{\ddot{x}} = \sqrt{2 \ln(n_1 T)} + \frac{0.5772}{\sqrt{2 \ln(n_1 T)}} \quad (\text{C26.11-5})$$

where T = length of time over which the minimum acceleration is computed, usually taken to be 3,600 s to represent 1 h.

Example calculations of maximum along-wind displacement, rms along-wind acceleration, and maximum along-wind acceleration are given in Table C26.11-2.

Approximate Fundamental Frequency. To estimate the dynamic response of structures, knowledge of the fundamental frequency (lowest natural frequency) of the structure is essential. This value would also assist in determining whether the dynamic

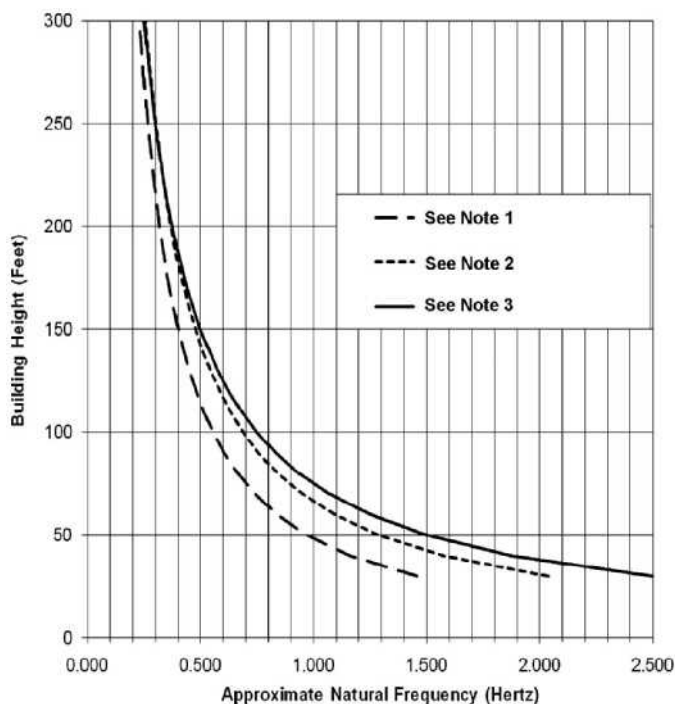
Table C26.11-2. Example Calculation

Example Calculation – Along-wind Response See Table C26.9-1 for additional items not shown		
Item	Value	Source
V Basic wind speed	115 mph	Figure 26.5-1A
ρ Air density	0.0024 slug/ft ³	Site elevation near sea level
C_{pw} External pressure coefficient, windward wall	0.8	Figure 27.4-1
C_{pw} External pressure coefficient, leeward wall	−0.8	Figure 27.4-1
C_{fx} Along-wind force coefficient	1.3	
ξ Mode shape power law exponent	1	Analysis of structure
α Peak velocity power law exponent	1/7	Table 26.9-1
$\hat{b}$ Velocity profile parameter	0.84	Table 26.9-1
$\hat{V}_z$ 3-s gust velocity at height $\bar{z}$	199 ft/s	(C26.9-1a)
K Modal load parameter	0.501	
ρ_b Building density	12 lbm/ft ³ , 0.3727 slug/ft ³	Building design
μ Building mass per unit height	3727 slug/ft	
m_1 Modal mass	745,400 slug	
MAXIMUM ALONG-WIND DISPLACEMENT AT BUILDING TOP		
$\phi(h)$ Mode shape at $z = h$	1.0	
$X_{\max}(h)$	1.86 ft	(C26.9-1)
RMS ALONG-WIND ACCELERATION AT BUILDING TOP		
V Basic wind speed, 10-yr MRI	76 mph	Figure CC-1
$\bar{V}_z$ Mean velocity at height $\bar{z}$	91.2 ft/s	(C26.9-3)
R Resonant response factor	0.755	(26.9-12)
$\sigma_{\ddot{x}}(h)$	0.135 ft/s ²	(C26.9-2)
MAXIMUM ALONG-WIND ACCELERATION AT BUILDING TOP		
T Time period for maximum	3600 s	Traditional
$g_{\ddot{x}}$ Peak factor	3.79	(C26.9-5)
$\ddot{X}_{\max}(h)$	0.512 ft/s ²	(C26.9-4)

response estimates are necessary. Most computer codes used in the analysis of structures would provide estimates of the natural frequencies of the structure being analyzed. For the preliminary design stages, some empirical relationships for building period T_a ($T_a = 1/n_1$) are available in the earthquake-related chapters of this standard. However, these expressions are based on recommendations for earthquake design with inherent bias toward higher estimates of fundamental frequencies (Goel and Chopra 1997, 1998). For wind design applications, these values may be unconservative because an estimated frequency higher than the actual frequency would yield lower values of the gust-effect factor and thus a lower design wind pressure. However, Goel and Chopra (1997, 1998) also cite lower bound estimates of frequency that are more suited for use in wind applications and are now given in Section 26.11.2; graphs of these expressions are shown in Fig. C26.11-1.

Because these expressions are based on regular buildings, limitations based on height and slenderness are required. The effective length, L_{eff} , uses a height-weighted average of the along-wind length of the building for slenderness evaluation. The top portion of the building is most important; hence, the height-weighted average is appropriate. This method is an appropriate first-order equation for addressing buildings with setbacks. Explicit calculation of the gust-effect factor per the other methods given in Section 26.11 can still be performed.

Observations from wind tunnel testing of buildings where frequency is calculated using analysis software show that the



Notes

1. Eq. (26.11-2): $22.2/h^{0.8}$
2. Eq. (26.11-3): $43.5/h^{0.9}$
3. Eq. (26.11-4): $75/h$

FIGURE C26.11-1 Equations for Approximate Lower-bound Natural Frequency n_a versus Building Height

following expression for frequency can be used for steel and concrete buildings less than about 400 ft (122 m) in height:

$$n_1 = 100/H(\text{ft}) \text{ average value} \quad (\text{C26.11-6})$$

$$n_a = 75/H(\text{ft}) \text{ lower bound value} \quad (\text{C26.11-7})$$

Eq. (C26.11-7) for the lower bound value is provided in Section 26.11.3.

Based on full-scale measurements of buildings under the action of wind, the following expression has been proposed for wind applications (Zhou and Kareem 2001a; Zhou et al. 2002):

$$f_{n1} = 150/h \text{ (ft)} \quad (\text{C26.11-8})$$

This frequency expression is based on older buildings and overestimates the frequency common in U.S. construction for smaller buildings less than 400 ft (122 m) in height, but it becomes more accurate for tall buildings greater than 400 ft (122 m) in height.

Studies in Japan involving a suite of buildings under low-amplitude excitations have led to the following expressions for natural frequencies of buildings (Sataka et al. 2003):

$$n_1 = 220/h \text{ (ft)} \text{ (concrete buildings)} \quad (\text{C26.11-9})$$

$$n_1 = 164/h \text{ (ft)} \text{ (steel buildings)} \quad (\text{C26.11-10})$$

These expressions result in higher frequency estimates than those obtained from the general expression given in Eqs. (C26.11-6) through (C26.11-8), particularly since the Japanese data set has limited observations for the more flexible buildings sensitive to wind effects, and Japanese construction tends to be stiffer.

For cantilevered masts or poles of uniform cross section (in which bending action dominates)

$$n_1 = (0.56/h^2)\sqrt{EI/m}$$

where EI is the bending stiffness of the section and m is the mass/unit height. This formula may be used for masts with a slight taper, using average value of EI and m (ECCS 1978).

An approximate formula for cantilevered, tapered, circular poles (ECCS 1978) is

$$n_1 \approx [\lambda/(2\pi h^2)]\sqrt{EI/m} \quad (\text{C26.11-12})$$

where h is the height of the pole, and E , I , and m are calculated for the cross section at the base. λ depends on the wall thicknesses at the tip and base, e_t and e_b , and external diameter at the tip and base, d_t and d_b , according to the following formula:

$$\lambda = \left[1.9 \exp\left(\frac{-4d_t}{d_b}\right) \right] + \left[\frac{6.65}{0.9 + (e_t/e_b)^{0.666}} \right] \quad (\text{C26.11-13})$$

Eq. (C26.11-12) reduces to Eq. (C26.11-11) for uniform masts. For freestanding lattice towers (without added ancillaries, such as antennas or lighting frames) (Standards Australia 1994):

$$n_1 \approx 1,500w_a/h^2 \quad (\text{C26.11-14})$$

where w_a is the average width of the structure in meters and h is tower height. An alternative formula for lattice towers (with added ancillaries) (Wyatt 1984) is

$$n_1 = \left(\frac{L_N}{h}\right)^{2/3} \left(\frac{w_b}{h}\right)^{1/2} \quad (\text{C26.11-15})$$

where w_b = tower base width and $L_N = 270$ m for square base towers, or 230 m for triangular base towers.

Structural Damping. Structural damping is a measure of energy dissipation in a vibrating structure that results in bringing the structure to a quiescent state. The damping is defined as the ratio of the energy dissipated in one oscillation cycle to the maximum amount of energy in the structure in that cycle. There are as many structural damping mechanisms as there are modes of converting mechanical energy into heat. The most important mechanisms are material damping and interfacial damping.

In engineering practice, the damping mechanism is often approximated as viscous damping because it leads to a linear equation of motion. This damping measure, in terms of the damping ratio, is usually assigned based on the construction material, for example, steel or concrete. The calculation of dynamic load effects requires damping ratio as an input. In wind applications, damping ratios of 1% and 2% are typically used in the United States for steel and concrete buildings at serviceability levels, respectively, while ISO (1997) suggests 1% and 1.5% for steel and concrete, respectively. Damping ratios for buildings under ultimate strength design conditions may be significantly higher, and 2.5% to 3% is commonly assumed. Damping values for steel support structures for signs, chimneys, and towers may be much lower than buildings and may fall in the range of 0.15–0.5%. Damping values of special structures like steel stacks can be as low as 0.2–0.6% and 0.3–1.0% for unlined and lined steel chimneys, respectively (ASME 1992; CICIND 1999). These values may provide some guidance for design. Damping levels used in wind load applications are smaller than the 5% damping ratios common in seismic applications because buildings subjected to wind loads respond essentially elastically, whereas buildings subjected to design-level earthquakes respond inelastically at higher damping levels.

Because the level of structural response in the strength and serviceability limit states is different, the damping values

associated with these states may differ. Furthermore, because of the number of mechanisms responsible for damping, the limited full-scale data manifest a dependence on factors such as material, height, and type of structural system and foundation (Kijewski-Correa et al. 2013). The Committee on Damping of the Architectural Institute of Japan suggests different damping values for these states based on a large damping database described in Sataka et al. (2003).

Recently, the NatHaz group has developed an interactive database of full-scale experimentally determined modal damping ratios based on the database (Kareem et al. 2012). The database is publicly available at <https://vortex-winds.org> and is equipped with a query-based Web interface for the rapid identification of modal damping ratios that satisfy specific requirements, such as geometric form, structural system, construction material, foundation type, and building use. A recent publication offers a data-driven model of damping that has been validated with several full-scale studies (Spence and Kareem 2014).

In addition to structural damping, aerodynamic damping may be experienced by a structure oscillating in air. In general, the aerodynamic damping contribution is quite small compared to the structural damping, and it is positive in low to moderate wind speeds. Depending on the structural shape, at some wind velocities the aerodynamic damping may become negative, which can lead to unstable oscillations. In these cases, reference should be made to recognized literature or a wind tunnel study.

Alternate Procedure to Calculate Wind Loads. The concept of the gust-effect factor implies that the effect of gusts can be adequately accounted for by multiplying the mean wind load distribution with height by a single factor. This is an approximation. If a more accurate representation of gust effects is required, the alternative procedure in this section can be used. It takes account of the fact that the inertial forces created by the building's mass, as it moves under wind action, have a different distribution with height than the mean wind loads or the loads caused by the direct actions of gusts (Zhou and Kareem 2001a; Chen and Kareem 2004). The alternate formulation of the equivalent static load distribution uses the peak base bending moment and expresses it in terms of inertial forces at different building levels. A base bending moment, instead of the base shear as in earthquake engineering, is used for the wind loads because it is less sensitive to deviations from a linear mode shape. For a more detailed discussion on this wind-loading procedure, see Zhou and Kareem (2001a, 2001b) and Chen and Kareem (2004).

Alternate Procedure: Along-Wind Equivalent Static Wind Loading. The equivalent static wind loading for the mean, background, and resonant components is obtained using the procedure outlined in this section.

Mean wind load component, $\bar{P}_j$, is given by

$$\bar{P}_j = q_j \times C_p \times A_j \times \bar{G} \quad (\text{C26.11-16})$$

where

j = floor level;
 q_j = velocity pressure at height z_j ;
 z_j = height of the j th floor above the ground level;
 C_p = external pressure coefficient; and
 $\bar{G} = 0.925 \times (1 + 1.7g_v I_z)^{-1}$ = gust velocity factor.

Peak background wind load component, $\hat{P}_{Bj}$, at the j th floor level is given similarly by

$$\hat{P}_{Bj} = \bar{P}_j \times G_B / \bar{G} \quad (\text{C26.11-17})$$

where

$$G_B = 0.925 \times \left(\frac{1.7I_z \times g_Q Q}{1 + 1.7g_v I_z} \right)$$

is the background component of the gust-effect factor.

Peak resonant wind load component, $\hat{P}_{Rj}$, at the j th floor level is obtained by distributing the resonant base bending moment response to each level:

$$\hat{P}_{Rj} = C_{Mj} \hat{M}_R \quad (\text{C26.11-18})$$

$$C_{Mj} = \frac{w_j \phi_j}{\sum w_j \phi_j z_j} \quad (\text{C26.11-19})$$

$$\hat{M}_R = \bar{M} \cdot G_R / \bar{G} \quad (\text{C26.11-20})$$

$$\bar{M} = \sum_{j=1,n} \bar{P}_j \cdot z_j \quad (\text{C26.11-21})$$

where

C_{Mj} = vertical load distribution factor;

$\hat{M}_R$ = peak resonant component of the base bending moment response;

w_j = portion of the total gravity load of the building located or assigned to level j ;

ϕ_j = first structural mode shape value at level j ;

$\bar{M}$ = mean base bending produced by mean wind load;

$G_R = 0.925 \times [(1.7I_z \times g_R R) / (1 + 1.7g_v I_z)]$ resonant component of the gust-effect factor; and

n = total stories of the building.

Alternate Procedure: Along-Wind Response. Through a simple static analysis, the peak building response in the along-wind direction can be obtained by

$$\hat{r} = \bar{r} + \sqrt{\hat{r}_B^2 + \hat{r}_R^2} \quad (\text{C26.11-22})$$

where $\bar{r}$, $\hat{r}_B$, and $\hat{r}_R$ = mean, peak background, and resonant response components of interest, for example, shear forces, moment, or displacement. Once the equivalent static wind load distribution is obtained, any response component, including acceleration, can be obtained using a simple static analysis. It is suggested that caution must be exercised when combining the loads instead of response according to the preceding expression, for example:

$$\hat{P}_j = \bar{P}_j + \sqrt{\hat{P}_{Bj}^2 + \hat{P}_{Rj}^2} \quad (\text{C26.11-23})$$

because the background and the resonant load components have normally different distributions along the building height. Additional background can be found in Zhou and Kareem (2001b); Zhou et al. (2002), and Chen and Kareem (2004).

Aerodynamic Loads on Tall Buildings: An Interactive Database. Under the action of wind, tall buildings oscillate simultaneously in the along-wind, across-wind, and torsional directions. While the along-wind loads have been successfully treated in terms of gust loading factors based on quasi-steady and strip theories, the across-wind and torsional loads cannot be treated in this manner because these loads cannot be related in a straightforward manner to fluctuations in the approach flow. As a result, most current codes and standards provide little guidance for the across-wind and torsional response (Zhou et al. 2002; Chen and Kareem 2004; Kwon and Kareem 2013; Bernardini et al. 2013).

To provide some guidance at the preliminary design stages of buildings, an interactive aerodynamic loads database for

Table C26.11-3. Along-Wind, Across-Wind, Torsional Moments, and Acceleration Response

Load Components	σ_{CM}	Survivability Design			Serviceability Design				
		Aerodynamic load coefficient		Base Moments (10 ⁶ kips-ft)	Aerodynamic load coefficient		Accelerations (milli-g or rad/s ²)		
				Corner					
f_1	$C_M(f_1)$	M	f_1	$C_M(f_1)$	σ_a	X	Y		
ASCE 7-10	—	—	—	1.73	—	—	1.95	2.77	3.24
Along-wind	0.109	0.193	0.046	1.72	0.292	0.022	2.03		
Across-wind	0.133	0.193	0.093	1.82	0.292	0.024	2.64		
Torsional	0.044	0.337	0.040	0.086	0.512	0.043	0.0001		

Note: As this database is experimental in nature thus has limitation in scope, it can be conveniently expanded using additional data as it becomes available.

assessing dynamic wind-induced loads on a suite of generic isolated buildings is introduced (Zhou et al. 2002; Kwon et al. 2008). Although the analysis based on this experimental database is not intended to replace wind tunnel testing in the final design stages, it provides users a methodology to approximate the previously untreated across-wind and torsional responses in the early design stages. The database consists of high-frequency base balance measurements involving seven rectangular building models, with side ratio (D/B , where D is the depth of the building section along the oncoming wind direction) from 1/3 to 3, and three aspect ratios for each building model in two approach flows, namely, BL_1 ($\alpha = 0.16$) and BL_2 ($\alpha = 0.35$) corresponding to an open and an urban environment, respectively. The data are accessible with a user-friendly, Java-based Internet applet, the NatHaz Aerodynamic Loads Database, version 2.0 (NALD 2012). Through the use of this interactive portal, users can select the geometry and dimensions of a model building from the available choices and specify an urban or suburban condition. Upon doing so, the aerodynamic load spectra for the along-wind, across-wind, or torsional directions are displayed with a Java interface permitting users to specify a reduced frequency (building frequency $\times$ building dimension/wind velocity) of interest and automatically obtain the corresponding spectral value. When coupled with the supporting Web documentation, examples, and concise analysis procedure, the database provides a comprehensive tool for computation of wind-induced response of tall buildings, suitable as a design guide in the preliminary stages.

Example: An example tall building is used to demonstrate the analysis using the database. The building is a square steel tall building with size $H \times W_1 \times W_2 = 656 \times 131 \times 131$ ft ($200 \times 40 \times 40$ m) and an average radius of gyration of 59 ft (18 m).

The three fundamental mode frequencies, f_1 , are 0.2, 0.2, and 0.35 Hz in X , Y , and Z directions, respectively; the mode shapes are all linear, or β is equal to 1.0, and there is no modal coupling. The building density is equal to 0.485 slugs/ft³ (250 kg/m³). This building is located in Exposure A or close to the BL_2 test condition of the Internet-based database (Zhou et al. 2002; Kwon et al. 2008). In this location, the 3-s design gust speed at a 700-year recurrence interval is 115 mi/h (51 m/s) in ASCE 7-16. For serviceability requirements, 3-s design gust speed with 10-year MRI is equal to 76 mi/h (34 m/s) in ASCE 7-16. For the sake of illustration only, the first-mode critical structural damping ratio, ζ_1 , is to be 0.01 for both survivability and serviceability design.

Using these aerodynamic data and the procedures provided in Zhou et al. (2002) and Kwon et al. (2008), the wind-load effects are evaluated and the results are presented in Table C26.11-3. This table includes base moments and acceleration response in the along-wind direction obtained by the procedure in

ASCE 7-16. It should be pointed out that the building experiences higher across-wind load effects when compared to the along-wind response for this example, which reiterates the significance of wind loads and their effects in the across-wind direction and the need for such database-enabled design approaches.

C26.12 ENCLOSURE CLASSIFICATION

The magnitude and sense of internal pressure are dependent upon the magnitude and location of openings around the building envelope with respect to a given wind direction. Accordingly, the standard requires that a determination be made of the amount of openings in the envelope to assess enclosure classification (enclosed, partially enclosed, or open). *Openings* are specifically defined in this version of the standard as apertures or holes in the building envelope which allow air to flow through the building envelope and which are designed as “open” during design winds. Examples include doors, operable windows, air intake exhausts for air conditioning and/or ventilation systems, gaps around doors, deliberate gaps in cladding, and flexible and operable louvers. The porosity for an “enclosed building” is such that there are not sufficient openings in the exterior building envelope to allow significant air flow into the building. The porosity of a “partially enclosed” building is such that there are sufficient openings in the building envelope windward wall to allow for wind to enter the building; however, there are not sufficient openings in the remaining portions of the building envelope to allow air flow out of the building without a buildup of internal pressure. The porosity for a “partially open” building is such that there exist sufficient openings in the building envelope windward wall to allow for air flow into the building and sufficient openings exist in the remaining portions of the building envelope to allow for some air flow out of the building but with some buildup of internal pressure. The porosity for the “open building” is such that air flow can enter and exit the building without a significant buildup of internal pressure. The classification of a “partially open” building has been added to the standard to help the user in the understanding that a building with openings and significant porosity (such as an open parking garage, for example) that does not meet the requirements of the “partially enclosed” classification does not automatically classify the building as “open” or “enclosed.” Once the enclosure classification is known, the designer enters Table 26.13-1 to select the appropriate internal pressure coefficient.

This version of the standard has four terms applicable to enclosure: wind-borne debris regions, glazing, impact-resistant glazing, and impact-protective system. *Wind-borne debris regions* are specified to alert the designer to areas requiring consideration

of missile-impact design and potential openings in the building envelope. *Glazing* is defined as any glass or transparent or translucent plastic sheet used in windows, doors, skylights, or curtain walls. *Impact-resistant glazing* is specifically defined as glazing that has been shown by testing to withstand the impact of test missiles. *Impact-protective systems* over glazing can be shutters or screens designed to withstand wind-borne debris impact. Impact resistance of glazing and protective systems can be tested using the test method specified in ASTM E1886, with missiles, impact speeds, and pass/fail criteria specified in ASTM E1996. Glazing in sectional garage doors and rolling doors can be tested for impact resistance with missiles, impact speeds, and pass-fail criteria specified in ANSI/DASMA 115. Other methods involving opening protection of building envelope systems are acceptable when approved by the Authority Having Jurisdiction. Origins of missile-impact provisions contained in these standards are summarized in Minor (1994) and Twisdale et al. (1996).

Section 26.12.3 requires glazing in Category II, III, and IV buildings in wind-borne debris regions to be protected with an impact-protective system or to be made of impact-resistant glazing to reduce the amount of wind and water damage to buildings during design windstorm events.

The standard requires all glazing in the lower 60 ft (18.3 m) of Category II, III, or IV buildings sited in wind-borne debris regions to be protected with an impact-protective system or to be made of impact-resistant glazing. Glazing higher than 60 ft (18.3 m) above grade may be broken by wind-borne debris when a debris source is present, such as aggregate-surfaced roofs on buildings within 1,500 ft (457 m) of the new building. This includes gravel or stone used as ballast that is not protected by a sufficiently high parapet. Accordingly, the glazing in the new building, from 30 ft (9.1 m) above the source building to grade needs to be protected with an impact-protective system or be made of impact-resistant glazing. If loose roof aggregate is proposed for the new building, it too should be considered as a debris source because aggregate can be blown off the roof and be propelled into glazing on the leeward side of the building. Although other types of wind-borne debris can impact glazing higher than 60 ft (18.3 m) above grade, at these higher elevations, loose roof aggregate has been the predominant debris source in previous wind events. The requirement for protection 30 ft (9.1 m) above the debris source is to account for debris that can be lifted during flight. The following references provide further information regarding debris damage to glazing: Beason et al. (1984), Minor (1985, 1994), Kareem (1986), and Behr and Minor (1994).

Although wind-borne debris can occur in just about any condition, the level of risk in comparison to the postulated debris regions and impact criteria may also be lower than that determined for the purpose of standardization. For example, individual buildings may be sited away from likely debris sources that would generate significant risk of impacts similar in magnitude to pea gravel (i.e., as simulated by 2-g steel balls in impact tests) or butt-on 2×4 impacts as required in impact-testing criteria. This situation describes a condition of low vulnerability only as a result of limited debris sources within the vicinity of the building. In other cases, potential sources of debris may be present, but extenuating conditions can lower the risk. These extenuating conditions include the type of materials and surrounding construction, the level of protection offered by surrounding exposure conditions, and the design wind speed. Therefore, the risk of impact may differ from those postulated as a result of the conditions specifically enumerated in the standard and the referenced impact standards. There are vastly differing opinions regarding the significance of these parameters that are not fully considered in developing standardized debris regions or referenced impact criteria.

The definition of the wind-borne debris regions for Risk Category II buildings and structures was chosen such that the coastal areas included in the wind-borne debris regions are approximately consistent with those given in ASCE 7-05 and prior editions. Thus, the new wind speed contours that define the wind-borne debris regions in Section 26.12.3.1 are not direct conversions of the wind speed contours that are defined in ASCE 7-05, as shown in Table C26.5-7.

While the coastal areas included in the wind-borne debris regions for Risk Category II are approximately consistent with those given in ASCE 7-05, significant reductions in the area of wind-borne debris regions for this Risk Category occur around Jacksonville, Florida, in the Florida Panhandle, and inland from the coast of North Carolina.

The introduction in ASCE 7-10 of separate maps for different Risk Categories provides a means for achieving a more risk-consistent approach for defining wind-borne debris regions. The approach selected was to link the geographical definition of the wind-borne debris regions to the wind speed contours in the maps that correspond to the particular Risk Category, resulting in expansion of the wind-borne debris region for some Risk Category III and all Risk Category IV buildings and structures. A review of the types of buildings and structures currently included in Risk Category III suggests that in the expanded wind-borne debris region life-safety issues would be most important for health-care facilities. Consequently, the committee chose to apply the expanded wind-borne debris protection requirement to this type of Risk Category III facilities and not to all Risk Category III buildings and structures.

C26.13 INTERNAL PRESSURE COEFFICIENTS

The internal pressure coefficient values in Table 26.13-1 were obtained from wind tunnel tests (Stathopoulos et al. 1979) and full-scale data (Yeatts and Mehta 1993). Even though the wind tunnel tests were conducted primarily for low-rise buildings, the internal pressure coefficient values are assumed to be valid for buildings of any height. The values (GC_{pi}) = +0.18 and -0.18 are for enclosed buildings. It is assumed that the building has no dominant opening or openings and that the small leakage paths that do exist are essentially uniformly distributed over the building's envelope. The internal pressure coefficient values for partially enclosed buildings assume that the building has a dominant opening or openings. For such a building, the internal pressure is dictated by the exterior pressure at the opening and is typically increased substantially as a result. Net loads (i.e., the combination of the internal and exterior pressures) are therefore also significantly increased on the building surfaces that do not contain the opening. Therefore, higher (GC_{pi}) values of +0.55 and -0.55 are applicable to this case. These values include a reduction factor to account for the lack of perfect correlation between the internal pressure and the external pressures on the building surfaces not containing the opening (Irwin 1987; Beste and Cermak 1996). Taken in isolation, the internal pressure coefficients can reach values of ± 0.8 (or possibly even higher on the negative side).

For partially enclosed buildings containing a large unpartitioned space, the response time of the internal pressure is increased, and this increase reduces the ability of the internal pressure to respond to rapid changes in pressure at an opening. The gust factor applicable to the internal pressure is therefore reduced. Eq. (26.13-1), which is based on Vickery and Bloxham (1992) and Irwin and Dunn (1994), is provided as a means of adjusting the gust factor for this effect on structures with large internal spaces, such as stadiums and arenas.

Because of the nature of hurricane winds and exposure to debris hazards (Minor and Behr 1993), glazing located below

60 ft (18.3 m) above the ground level of buildings sited in wind-borne debris regions has a widely varying and comparatively higher vulnerability to breakage from missiles, unless the glazing can withstand reasonable missile loads and subsequent wind loading, or the glazing is protected by suitable shutters. [See Section C26.12 for discussion of glazing above 60 ft (18.3 m).] When glazing is breached by missiles, development of higher internal pressure may result, which can overload the cladding or structure if the higher pressure was not accounted for in the design. Breaching of glazing can also result in a significant amount of water infiltration, which typically results in considerable damage to the building and its contents (Surry et al. 1977; Reinhold 1982; Stubbs and Perry 1993).

The influence of compartmentalization on the distribution of increased internal pressure has not been researched. If the space behind breached glazing is separated from the remainder of the building by a sufficiently strong and reasonably airtight compartment, the increased internal pressure would likely be confined to that compartment. However, if the compartment is breached (e.g., by an open corridor door or by collapse of the compartment wall), the increased internal pressure will spread beyond the initial compartment quite rapidly. The next compartment may contain the higher pressure, or it too could be breached, thereby allowing the high internal pressure to continue to propagate. Because of the great amount of air leakage that often occurs at large hangar doors, designers of hangars should consider using the internal pressure coefficients in Table 26.13-1 for partially enclosed buildings

C26.14 TORNADO LIMITATION

Tornadoes have not been considered in the wind load provisions because of their very low probability of occurrence. However, some building owners might want to have portions of their buildings designed to provide a greater level of occupant protection from tornadoes or minimize building damage associated with EF0–EF2 rated tornadoes (see Section C26.14.1 for tornado ratings). Maintaining continuity of building operations for facilities such as emergency operations centers and hospitals is also a factor when considering whether or not to design for tornado wind and wind-borne debris loads. This section provides information and design guidance for those designers and building owners who desire to design for reduced property damage or increased occupant protection, in case the building under consideration is impacted by a tornado.

The following topics are addressed in this section: 1) tornado wind speeds and probabilities, 2) wind pressures induced by tornadoes versus other windstorms, 3) designing for occupant protection, 4) designing to minimize building damage, 5) designing to maintain continuity of building operations, and 6) designing trussed communications towers for wind-borne debris. (Mehta et al. (1976), Minor (1982), Minor et al. (1977), and Wen and Chu (1973) provide early background information on tornadoes.)

C26.14.1 Tornado Wind Speeds and Probabilities. The National Weather Service (NWS) rates tornado severity according to the six levels of observed damage in the Enhanced Fujita Scale (EF Scale). The scale ranges from EF0 to EF5. See Table C26.14-1 for the wind speeds associated with the EF ratings. Damage indicators (DI) and degrees of damage (DOD) are used to establish EF ratings. DIs consist of buildings, other structures, and trees. Information on the EF scale and a description of DIs and DODs can be found in McDonald and Mehta (2006).

NWS data indicate that the median number of tornadoes in the United States between 1990 and 2014 was 1,173 per year. During this time, the fewest number of tornadoes (888) occurred in 2014,

Table C26.14-1 Enhanced Fujita (EF) Scale

EF Number	Wind Speed (mph)	(m/s)
EF0	65–85	29–38
EF1	86–110	39–49
EF2	111–135	50–60
EF3	136–165	61–73
EF4	166–200	74–89
EF5	>200	>89

Note: Speeds are for 3-s peak gust, Exposure C, 33 ft (10 m) above grade. Conversion of mph to m/s: $\text{mph} \times 0.44704 = \text{m/s}$.

Source: NOAA (<http://www.spc.noaa.gov/efscale/ef-scale.html>).

and the greatest number (1,817) occurred in 2004. Tornado-related winds have a significantly lower probability of occurrence at a specific location than the high winds associated with meteorological events (frontal systems, thunderstorms, and hurricane winds) responsible for the basic wind speeds given in ASCE 7. The probability of occurrence is a function of the area covered by a tornado and of the specific location. The probability of a site-specific EF0 to EF1 rated tornado strike in the central portion of the United States is on the order of a 4,000-year MRI (Ramsdell and Rishel 2007). (Considering that Risk Category IV buildings and structures have an MRI of 3,000 years in the standard, designing for an EF1 tornado would only result in a small increase in design wind pressures. It would be prudent to design for an EF1 or greater tornado, as discussed in Sections C26.14.4 and C26.14.5.) In the areas of the country where the risk of EF4 and EF5 rated tornadoes is greatest, the annual probabilities that a particular building will be affected by an EF4 or EF5 rated tornado are on the order of 10^{-7} (a 10,000,000-year MRI) (Ramsdell and Rishel 2007). Tornadoes in the West are rare, as illustrated by NWS annual tornado maps from 1952 to 2011 (<http://www.spc.noaa.gov/wcm/annualtornadomaps/>).

Fig. C26.14-1 shows the recorded EF3–EF5 rated tornadoes between 1950 and 2013. Of the 56,221 recorded tornadoes that occurred between 1950 and 2011, 95% were rated as EF0–EF2 by the NWS, 4% were rated as EF3, and 1% were rated as EF4–EF5.

Damage investigations have indicated that tornado winds are more likely to generate more wind-borne debris compared to non-tornadic winds of the same speed. EF0 and EF1 rated tornadoes may generate wind-borne debris that can break unprotected glazing and puncture many types of door, wall, and roof assemblies.

As illustrated in Table C26.14-3, depending on a building's location and Risk Category, EF2 and EF3 rated tornadoes produce wind pressures that range from below to above those derived from ASCE 7-10 for hurricane-prone regions. Hence, for buildings designed for wind pressure in accordance with ASCE 7, the performance of structural elements (i.e., MWFRS), doors, and walls in tornadoes will depend on the relationship between the tornado severity and the basic wind speed. For example, a building in Miami is expected to have greater resistance to strong tornadoes than a building in Orlando, where the basic wind speed is lower than Miami's basic wind speed. However, wind-borne debris can break unprotected glazing and puncture many types of door, wall, and roof assemblies. Even if the glazing is protected from hurricane debris, debris from an EF3 rated tornado may penetrate the glazing because the momentum of debris generated by an EF3 rated tornado may significantly exceed the impact test criteria adopted for hurricane opening protection.

EF3 through EF5 rated tornadoes can produce wind pressures and wind-borne debris loads that are in excess of those derived from the highest design wind speeds for hurricane-prone areas and the wind-borne debris test standards for opening protection

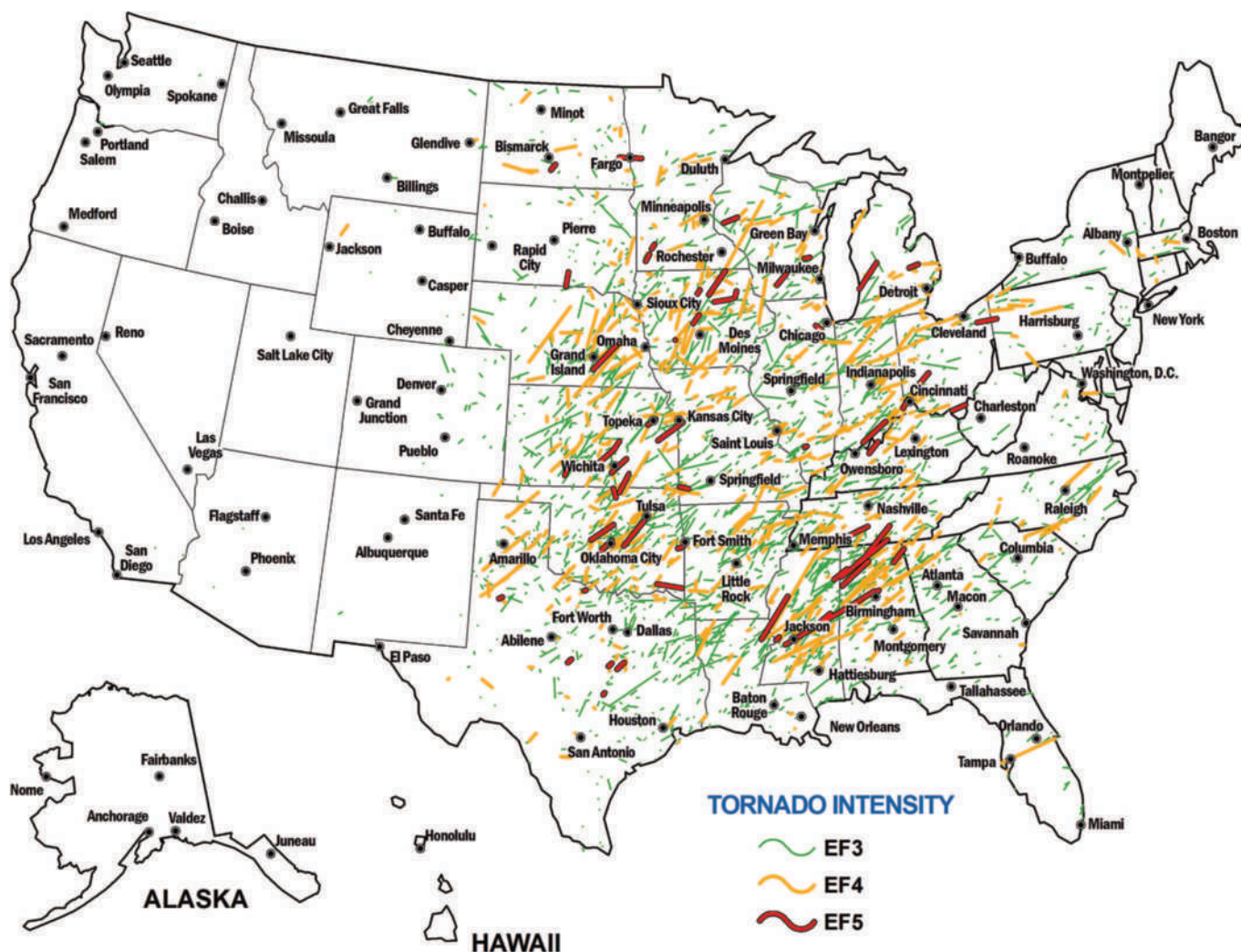


FIGURE C26.14-1 Recorded EF3–EF5 Rated Tornadoes, 1950–2013

Source: FEMA (2015), from National Oceanic and Atmospheric Administration, National Weather Service, Storm Prediction Center

in hurricane prone regions. Fig. C26.14-2 shows the design wind speed for tornado safe rooms recommended by FEMA P-361 (FEMA 2015). This wind speed map is the same as the ICC 500 tornado hazard map. These wind speeds likely have a probability of occurrence that is on the order of 1×10^{-6} to 1×10^{-7} per year (Ramsdell and Rishel 2007).

C26.14.2 Wind Pressures Induced by Tornadoes Versus Other Windstorms. Photo analysis of wind-borne debris shed from buildings indicates that tornado debris has a greater vertical trajectory than hurricane debris. Such observations and tornado simulations suggest that updrafts are greater in tornadoes than in other windstorms. Research by Mishra et al. (2008) indicates that atmospheric pressure drops play a significant role in wind pressures experienced in tornadoes. Laboratory research conducted in a tornado simulator (Haan et al. 2010) measured wind pressures on a gable roof of a residential-size building model. Measured pressures included the pressure drop in the simulated vortex and point pressures, which were integrated to determine MWFRS loads. Results have been compared with loads produced by both ASCE 7 directional and envelope MWFRS wind load provisions. The largest lateral loads determined from the tornado simulator test results are slightly lower than the lateral loads produced using the MWFRS directional method of ASCE 7, Chapter 27, but they are

up to about 50% larger than those produced using the MWFRS envelope method of ASCE 7, Chapter 28. For a tightly sealed building where the internal pressure in the building does not rapidly equalize with the atmospheric pressure drop in the tornado vortex, the uplift loads on the building obtained from the simulator tests can be substantially larger than those obtained from either the directional or envelope MWFRS methods of ASCE 7. However, at the other extreme, if an open building is assumed and the atmospheric pressure drop in the vortex is removed from the surface pressures on the building roof, the net uplift MWFRS loads from the simulator tests produces lower uplift loads on the building than those produced using the ASCE 7 directional and envelope MWFRS methods. In consideration of the typical leakage of most buildings, the use of internal pressure coefficients for a partially enclosed building and the fact that the building would have already experienced the highest horizontal winds in the tornado vortex before it was exposed to the largest pressure drop, which occurs in the vortex core, the Wind Load Subcommittee decided that the uplift load coefficients should not be further increased for most buildings to account for atmospheric pressure drop effects in the tornado vortex. This would not be an appropriate decision for nuclear power plant containment structures or other structures where extreme measures are taken to ensure that the interior of the building is sealed from atmospheric conditions.

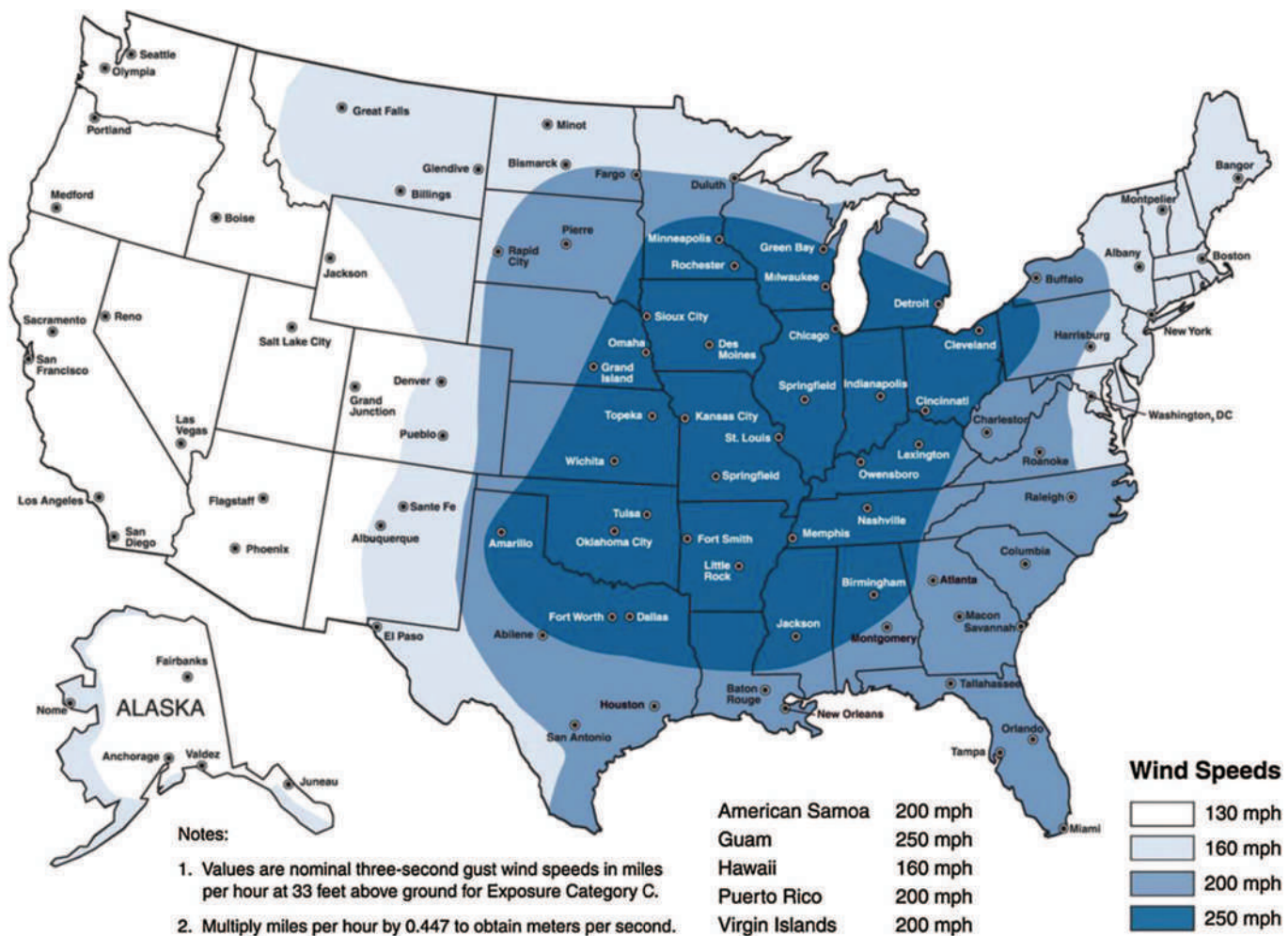


FIGURE C26.14-2 Tornado Safe Room Design Wind Speed Map

Source: FEMA (2008)

C26.14.3 Occupant Protection. A report (CDC 2012) on fatalities during the April 25–28, 2011, tornado outbreak illustrates the importance of providing tornado protection for building occupants. There were 351 recorded tornadoes during the outbreak, where 338 fatalities were caused by 27 of the tornadoes. According to the report, 90% of the people who died were in buildings at the time the tornadoes struck.

ICC 500 provides requirements for design and construction of residential and community storm shelters. FEMA P-320 (2014) provides prescriptive solutions for residential and small business safe rooms that will accommodate up to 16 occupants. FEMA P-361 (2015) provides comprehensive guidance for the design of community and residential safe rooms, as well as for quality assurance and quality control for their design and construction. The criteria in FEMA P-361 and ICC 500 are quite similar. All safe room criteria in FEMA P-361 meet the storm shelter requirements in ICC 500, but a few design and performance criteria in FEMA P-361 are more restrictive than those in ICC 500. The FEMA P-320 and P-361 documents provide additional guidance on design standards that should be followed in order for shelters to potentially qualify for federal grant funding.

Tornado shelters complying with ICC 500 or safe rooms complying with FEMA P-320 (2014) or P-361 (2015) are the recommended methods of protecting occupants from EF3, EF4, or EF5 rated tornadoes. However, if a tornado shelter or safe

room is not provided, some level of occupant protection may be achieved if the building is designed to minimize damage (see Section C26.14.4). Depending upon the magnitude of the design enhancements, the quality of construction, maintenance and repair of the building, the potential for roof structure and wall collapse is reduced if the building is struck by an EF0 to EF2 rated tornado. Hence, in addition to minimizing building damage, there will be some level of occupant protection, although the amount of protection is difficult to quantify. The design objective of minimizing damage and providing for occupant protection is discussed in van de Lindt et al. (2013). However, if occupants of such buildings do not have access to a shelter or safe room, they are exposed to a slight risk of death or injury from wind-borne debris generated by EF0 to EF2 rated tornadoes. They are also clearly susceptible to death or injury if the building is struck by a stronger tornado (EF3 to EF5).

For buildings that do not have an ICC 500-compliant storm shelter, or a FEMA P-320 (2014) or P-361 (2015) compliant safe room, FEMA P-431 (2009) provides guidance in identifying best available refuge areas.

C26.14.4 Minimizing Building Damage. Table C26.14-2 shows the comparison between hurricane and tornado wind speeds. By using design strategies consistent with construction in hurricane-prone regions, new buildings can be relatively econo-

Table C26.14-2 Comparison of Hurricane and Tornado Wind Speeds

Hurricane Category	Wind Speed over Land		EF Rating	Wind Speed	
	(mi/h)	(m/s)		(mi/h)	(m/s)
1	81–105	36–46	EF0	65–85	29–38
2	106–121	47–54	EF1	86–110	39–49
3	122–143	55–64	EF2	111–135	50–60
4	144–171	65–76	EF3	136–165	61–73
5	>171	>76	EF4	165–200	74–89
			EF5	>200	>89

Note: Speeds are 3-s peak gust, Exposure C, 33 ft (10 m) above grade. Conversion of mph to m/s: $\text{mph} \times 0.44704 = \text{m/s}$.

mically designed and built to be more resistant to most tornadoes (i.e., EF0–EF2) (Prevatt et al. 2012). Furthermore, published tornado simulator results suggest that the loads on the buildings may be somewhat similar to those from straight-line winds provided the tornado is large enough to engulf the building. With appropriate strengthening and selection of building materials and systems, the cost of tornado repairs can be reduced. However, when designing to minimize tornado damage, it is important to realize that wind-borne debris is likely to cause breaches in the building envelope. If exterior glazing is breached, interior ceilings and/or walls may collapse from wind pressures, and water damage is possible.

If a building is designed to minimize tornado damage, it is recommended that consideration also be given to including a tornado shelter complying with ICC 500 or a safe room complying with FEMA P-320 (2014) or P-361 (2015) for occupant protection. For example, if a medical research lab is designed for EF3 pressures in order to minimize building damage, a portion of the building could also be designed as a shelter to provide occupant protection in the event that the facility was struck by significant tornado-generated debris or winds from an EF4 or EF5 rated tornado.

Tornado Wind Pressure Calculations. There are several differences in the values of variables recommended for use in tornado wind pressure calculations as compared with normal wind pressure calculations. (The pressure calculation recommendations are consistent with those given in FEMA P-361 (2015) and ICC 500.) However, the methodology and the equations are the same as provided in Section 26.10.2, Eq. (26.10-1) (for velocity pressure), Section 27.3.1, Eq. (27.3-1) (for MWFRS), and Section 30.3.2 and Eq. 30.3-1 (for C&C). There are two methods presented in this commentary for the development of tornado wind pressures. Both methods achieve the same answer. The first method (Extended Method) is to use wind pressure calculation parameters (in equations noted above) modified for use in determining tornado pressures. Because this method does require some adjustment to almost every parameter, a second method (Simplified Method) is presented that reduces all of the changed parameters to a single multiplier. The Simplified Method is intended to provide a simple method of accounting for various tornado-related design considerations and for the uncertainties in tornado pressures caused by a lack of field pressure measurements and limited laboratory research.

The following provides a description and discussion of those differences that are then combined to provide a single factor, identified as a Tornado Factor or TF (i.e., the Simplified Method) that can be applied to results of ASCE 7 calculated pressures and wind loads to produce tornado design wind loads for a target tornado intensity. The TF factors shown in Table C26.14-4

reflect a combination of adjustment factors for internal pressure, gust factor, directionality and external pressure coefficient differences, and the uncertainty in these adjustment factors. Because the original design calculations may be based on Exposures B, C, or D and the tornado design pressures and loads assume Exposure C or D, the recommended TF factor is different depending on whether the original calculations were for Exposure Category B or for either Exposure Category C or D. Note that these factors are offered as a way to simplify the assessment of the increases in loads caused by tornado design, and the designer always has the option of using the Extended Method with the revised factors in the wind pressure equation suggested below for designing for tornado wind loads.

Extended Method. A tornado event is considered an ultimate wind speed event, so the tornado design wind speed for the upper end of the target design EF Scale should be used as the design wind speed. Wind speeds for tornado design may also come from ICC 500 or FEMA P-361 (2015). Because the tornado design wind speed is an ultimate wind speed, it is treated the same as the basic wind speeds provided in Fig. 26.5-1 (e.g., the load factor for strength design is 1.0 and the wind loads can be multiplied by 0.6 for use in allowable stress design).

It is recommended that the pressures always be determined based on an Exposure C condition except as noted following, regardless of the actual exposure of the site. It is anticipated that the winds of the tornado are going to create or at least resemble winds from an Exposure C terrain condition. Thus, even in more suburban settings, vegetation will be shredded and some roofs will be torn off, creating a smoother terrain than the original one. If the design exposure is D, the D exposure should also be used for tornado design.

It is recommended that the directionality factor be taken as $K_d = 1.0$ for a tornado. In Section C26.6, the following describes the reason for the directionality factor: “This factor accounts for two effects: (1) The reduced probability of maximum winds coming from any given direction, and (2) the reduced probability of the maximum pressure coefficient occurring for any given wind direction.” In the judgment of the Wind Load Subcommittee, it was decided to set $K_d = 1.0$ for wind pressures from tornadoes since neither of the two conditions used as the basis for $K_d = 0.85$ seemed to apply. Considering the rotational winds in a tornado, and the likelihood that at least one building corner or window will experience a coincident worst (GC_p) direction and a maximum or near maximum wind speed, a value of $K_d = 1.0$ seems prudent.

It is recommended that the topographic effect factor be taken as $K_{zt} = 1.0$, since topographic effects on tornadic wind speeds are not well understood.

The gust-effect factor G should be increased from 0.85 to 0.90 for rigid buildings. A $G = 0.90$ value is derived from the gust-effect factor used for rigid buildings of $G = 0.85$ without the 0.925 calibration factor incorporated within that value. Thus, for tornado design, $G = 0.85/0.925 = 0.92$, or it can be rounded down to 0.90. $G = 0.90$ should be the minimum gust-effect factor used.

Component and cladding (C&C) loads in ASCE 7 are based on (GC_p) values that are associated with a wind event that lasts for a period of time that is frequently taken as one hour. Most tornadoes move with a translational speed that limits the strongest winds to a matter of minutes, if not seconds. Consequently, a reduction in the C&C loads appears to be warranted (e.g., Kopp and Morrison 2011). Given the limited amount of research that has been conducted on wind pressures in tornado simulators, the Wind Load Subcommittee judgment is to limit the reduction to 10% until more information is available.

It is recommended that the internal pressure coefficient be taken as $(GC_{pi}) = \pm 0.55$. A breach of the building envelope is likely during a tornado, which will significantly increase the wind-induced internal pressure in the building while helping to promote equalization of atmospheric pressure drop in the vortex if it passes over the building. From the changes in design criteria noted above, the velocity pressure q for tornado pressure is

$$q = 0.00256 K_d K_z K_{zt} K_e V^2 \quad (\text{C26.14-1})$$

which is reduced to $0.00256 K_z V^2$, where V^2 is selected from the EF rating selected for design and K_z is based on Exposure C or D as appropriate.

Given the nature of the wind profile in a tornado, it is recommended that q be determined at the mean roof height, h , and that q_h be used throughout the pressure calculations as the value of velocity pressure, q .

For the MWFRS pressure, then

$$p = q_h GC_p - (\pm 0.55) \quad (\text{C26.14-2})$$

where

$G = 0.90$ or higher as determined by the current methods, and C_p = external pressure coefficient found by the current directional methods of Chapter 27.

For C&C pressures,

$$p = q_h [(0.9 \times (GC_p)) - (\pm 0.55)] \quad (\text{C26.14-3})$$

where (GC_p) is the external pressure coefficient for C&C found by current methods.

To summarize, the recommended tornado design wind pressures and loads for both MWFRS and C&C elements should be calculated using the following:

V = upper end of range of wind speed for target EF scale or the speed from ICC 500 or FEMA P-361 (2015) (Note: The EF scale does not provide an upper bound wind speed for EF5-rated tornadoes. When designing for EF5-rated tornadoes, the design wind speeds shown in Fig. C26.14-2 are reasonable upper bound speeds.);

K_z = velocity pressure exposure coefficient evaluated at mean roof height for Exposure C or D as appropriate;

$K_d = 1.0$;

$K_{zt} = 1.0$;

$K_e = 1.0$ or value from Table 26.9-1;

$(GC_{pi}) = \pm 0.55$;

$G(\text{MWFRS}) = 0.9$ or higher if appropriate from current methods;

$C_p(\text{MWFRS})$ = ASCE 7 values per Chapter 27 and directional analysis; and

$(GC_p)(\text{C&C})$ = ASCE 7 values for the appropriate zone.

It should be recognized that there is considerable uncertainty when it comes to the development and justification of tornado design wind loads. Nevertheless, it is also important to recognize that the EF scale for tornadoes is essentially a damage scale based on a series of damage indicators (DIs) and degrees of damage (DODs). The expert opinions that were used to establish the wind speeds associated with the DIs and DODs were at least partially shaped by observations of damage in hurricanes where there was a higher degree of confidence in wind speeds than in tornadoes. Consequently, changes in wind loads based on static pressure drop estimates in the middle of tornadoes is probably not

justified. Conversely, though, observations of dramatic structural failures in hurricanes are almost always accompanied by observations of breaching of the building envelope that would have produced significant internal pressures on the building or a part of the building. Because breaches in the building envelope are highly likely in tornadoes because of the amount of debris in the circulation and there is likely insufficient time to protect openings unless a passive system is used (which in turn might reduce the opportunity for pressure equalization), it is prudent to design for some measure of additional internal pressurization. This is done by setting the internal pressure coefficient to ± 0.55 .

Simplified Method. A series of calculations were carried out to estimate the effect of the various tornado design recommendations outlined above. These calculations are shown in Table C26.14-3, which illustrates quantitative comparisons of MWFRS pressures derived from ASCE 7-10 and estimated pressures induced by EF0–EF4 rated tornadoes. The calculations are based on a 30-ft by 30-ft enclosed building with a mean roof height of 22 ft and a gable roof angle of 35° sited in Exposure C, which is the building size and exposure used in research scaled up to a full-size building (Haan et al. 2010). The tornado calculations assume partially enclosed conditions (caused by broken glazing) and a Tornado Factor (TF) based on tornado design considerations outlined in this section. Based on these calculations of the increase in pressures from tornadoes on both walls and roofs, it was concluded that the use of simple single factors could be applied to the design wind loads calculated using normal ASCE 7 procedures and equations for enclosed buildings as a reasonable way for designers to address tornado design for their clients who wanted to evaluate options and to provide a measure of tornado damage reduction.

The C&C TF factors are conservatively based on the field zones of roofs and walls. These simple, singular factors (TF factors) are summarized in Table C26.14-4 and illustrated in Eq. (C26.14-4).

Note that these adjustments can be applied to either limit state “ultimate” design loads or to allowable stress design pressures or loads calculated for the building to arrive at tornado design pressures or loads.

Applying these TF factors from Table C26.14-4 follows:

$$p = q_i (GC_p - (GC_{pi})) \times TF \quad (\text{C26.14-4})$$

where the $q_i (GC_p - (GC_{pi}))$ portion of the equation is determined using the current ASCE 7 Chapter 27 directional procedure for

Table C26.14-3 MWFRS Pressure Comparisons – Straight Line Wind versus Tornado

	Wind Speed	Roof Uplift Pressure	Wall Negative Pressure
	mph	psf	psf
ASCE 7-16	115	–30	–22
ASCE 7-16	120	–33	–24
ASCE 7-16	180	–74	–55
ASCE 7-16	190	–82	–61
EF0 (upper end of range)	85	–27	–22
EF1 (upper end of range)	110	–45	–37
EF2 (upper end of range)	135	–68	–55
EF3 (upper end of range)	165	–102	–82
EF4 (upper end of range)	200	–150	–120

Note: Conversion of mph to m/s: $\text{mph} \times 0.44704 = \text{m/s}$; Conversion of psf to kN/m^2 : $\text{psf} \times 0.0479 = \text{kN/m}^2$

Table C26.14-4 Increases in Design Loads to Address Tornado Risks Using Recommended Tornado Factors

Original Enclosure Classification	Loading	Original Exposure B	Original Exposure C or D
Partially enclosed buildings	MWFRS	1.8	1.2
	C&C	1.6	1.1
Enclosed buildings ^a	MWFRS	2.5	1.6
	C&C	2.1	1.5

^aThe tornado factors to be used to increase the design loads on elements of enclosed buildings are based on the effects of high internal pressures. High internal pressures have a much greater effect on elements that typically receive less wind, so the net effect of these increase factors is typically much higher than would result if the building were designed for the specific tornado loads or if the tornado factors for partially enclosed buildings were used with partially enclosed building designs.

enclosed buildings. The wind speed used in Eq. (C26.14-4) to determine q is the selected design wind speed for the tornado design. Alternatively, the designer can calculate wind pressures for both MWFRS and C&C using the design wind speeds from ASCE 7, Fig. 26.5-1A, B, C, or D, or Fig. 26.5-2A, B, C, or D (or <http://windspeed.atcouncil.org/>), then scale the pressures for tornado design as follows:

$$p = q_i(GC_p - (GC_{pi})) \times (V_{\text{tornado}}/V_{\text{design}})^2 \times \text{TF}$$

where V_{tornado} is the selected tornado wind speed to be used for this tornado design and TF factors are from Table C16.14-4.

Example.

Description of Building

Building dimensions: 30 ft $\times$ 60 ft

Mean roof height = 40 ft

The building is located in Des Moines, Iowa, and is a two-story, light commercial building with a flat roof. The primary building use is computer server storage. The building is located in a commercial office park; the assumed exposure is B.

The owner has decided that he would like the building to be designed for an EF2 tornado. The top end of the EF2 tornado scale is 135 mph. The ASCE 7 wind speed for Risk Category II buildings in this location is 110 mph.

Determine the differences in wall and roof pressures (both MWFRS and C&C) for a tornado design for 135 mph compared to the design pressures using ASCE 7 mapped wind speeds.

Tornado wind pressure determination using the Extended Method

Wind speed $V = 135$ mph;

Mean roof height $h = 40$ ft;

$K_d = 1.0$;

$K_{zt} = 1.0$;

$K_z = 1.04$ (taken at Exposure C and 40 ft);

Velocity pressure $q = 0.00256K_zK_{zt}K_dV^2 = 0.00256 \times 1.04 \times 1.0 \times 1.0 \times 135^2 = 48.7$ psf; and

$G = 0.9$.

For the MWFRS: $p = q_h(GC_p - (GC_{pi}))$

Table C26.14-5 shows calculations for MWFRS wind pressures from the Extended Method.

Tornado wind pressure determination using the Single TF Factor Method

Wind speed $V = 135$ mi/h;

Mean roof height $h = 40$ ft;

$K_d = 0.85$;

$K_{zt} = 1.0$;

$K_z = 0.76$ (taken at Exposure B and 40 ft);

Velocity pressure $q = 0.00256K_zK_{zt}K_dV^2 = 0.00256 \times 0.76 \times 1.0 \times 0.85 \times 135^2 = 30.2$ psf; and $G = 0.85$.

For the MWFRS: $p = q_i(GC_p - (GC_{pi}))$

where (GC_{pi}) is for an enclosed condition and thus = ± 0.18 .

Table C26.14-6 shows MWFRS wind pressures using ASCE 7 parameters. Table C26.14-7 shows MWFRS wind pressures comparison: extended versus single-factor methods.

Multiplying by the MWFRS TF factor for an original B exposure = 2.5 (from Table C26.14-8).

Alternatively, the wind speeds can be scaled in accordance with Eq. (C26.14-5), where V_{tornado} is the tornado design wind speed (135 mph in this example) and V_{design} is the ASCE 7 mapped wind speed for the location (110 mph in this example).

Wind pressures p are developed using the ASCE 7 design wind speed and the ASCE 7 parameters:

$$(V_{\text{tornado}}/V_{\text{design}})^2 = (135/110)^2 = 1.51$$

The results using this method are shown in Table C26.14-8.

Load Path. It is necessary to take the pressures through the appropriate load paths so that connections between building elements can be designed. The most important load paths indicated by damage observations are the roof-to-wall connection, the wall-to-floor connection, the floor-to-foundation connection, any connection across floors, and the connection between the exterior walls at the corners. In addition to the exterior

Table C26.14-5 Example Problem MWFRS Wind Pressures from Extended Method

	q (psf)	C_p	$\pm(GC_{pi})$	Pressures with $-(GC_{pi})$	Pressures with $-(GC_{pi})$
MWFRS: Walls					
Windward	48.7	+0.8	± 0.55	61.8	8.3
Leeward	48.7	$L/B = 0.5, -0.5$	± 0.55	4.9	-48.7
		$L/B = 2.0, -0.3$		13.6	-39.9
Sidewalls	48.7	-0.7	± 0.55	-3.9	-57.5
MWFRS: Roof					
Wind perpendicular to 30-ft side	48.7	$h/L = 0.67$	± 0.55		
		$0-h, -0.9$		-12.7	-66.2
		$h-2h, -0.5$		4.9	-48.7
Wind perpendicular to 60-ft side	48.7	$h/L = 1.33$	± 0.55		
		$0-h/2, -1.3$		-30.2	-83.7
		$>h/2, -0.7$		-3.9	-57.5

Table C26.14-6 Example Problem MWFRS Wind Pressures Using ASCE 7 Parameters

	q (psf)	C_p	$\pm(GC_{pi})$	Pressures with $-(GC_{pi})$	Pressures with $+(GC_{pi})$
MWFRS: Walls					
Windward (at mean roof height)	30.2	+0.8	± 0.18	25.9	15.1
Leeward	30.2	$L/B = 0.5, -0.5$	± 0.18	-7.4	-18.2
		$L/B = 2.0, -0.3$		-2.3	-13.1
Sidewalls	30.2	-0.7	± 0.18	-12.5	-23.4
MWFRS: Roof					
Wind perpendicular to 30-ft side	30.2	$h/L = 0.67$	± 0.18		
		$0-h, -0.9$		-17.6	-28.5
		$h - 2h_i, -0.5$		-7.4	-18.2
Wind perpendicular to 60-ft side	30.2	$h/L = 1.33$	± 0.18		
		$0-h/2, -1.3$		-27.9	-38.7
		$>h/2, -0.7$		-12.5	-23.4

Table C26.14-7 Example Problem MWFRS Wind Pressures Comparison: Extended versus Single-Factor Methods

	Tornado Factor	ASCE 7 Enclosed Condition		Tornado Extended Method		Ratio of the Largest Pressures Single-Factor/ Extended
		Pressures with $-(GC_{pi})$	Pressures with $+(GC_{pi})$	Pressures with $-(GC_{pi})$	Pressures with $+(GC_{pi})$	
MWFRS: Walls						
Windward (at mean roof height)	2.5	$25.9 \times 2.5 = 64.9$	$15.1 \times 2.5 = 37.7$	61.8	8.3	1.05
Leeward (largest pressures)	2.5	$-7.4 \times 2.5 = -18.5$	$-18.2 \times 2.5 = -45.6$	4.9	-48.7	0.94
Sidewalls	2.5	$-12.5 \times 2.5 = -31.3$	$-23.4 \times 2.5 = -58.4$	-3.9	-57.5	1.02
MWFRS: Roof						
Wind perpendicular to 30-ft side (largest pressures)	2.5	$-17.6 \times 2.5 = -44.1$	$-28.5 \times 2.5 = -71.3$	-12.7	-66.2	1.08
Wind perpendicular to 60-ft side (largest pressures)	2.5	$-27.9 \times 2.5 = -69.8$	$-38.8 \times 2.5 = -96.9$	-30.2	-83.7	1.16

Table C26.14-8 Example Problem MWFRS Wind Pressures Comparison: Extended versus Single-Factor Methods with Factored Wind Speeds

	ASCE 7 Design Wind Speed		ASCE 7 Design Wind Speed × Correction × TF		Tornado Extended Method		Ratio of the Largest Pressures:Single- Factor/Extended
	Pressures with −(<i>GC_{pi}</i>)	Pressures with +(<i>GC_{pi}</i>)	Pressures with −(<i>GC_{pi}</i>)	Pressures with +(<i>GC_{pi}</i>)	Pressures with − <i>GC_{pi}</i>	Pressures with + <i>GC_{pi}</i>	
MWFRS: Walls							
Windward (at mean roof height)	17.2	10.0	$17.2 \times 2.5 \times 1.51 = 65.0$	$10.0 \times 2.5 \times 1.51 = 37.8$	61.8	8.3	1.05
Leeward (largest pressures)	−4.9	−12.1	$−4.9 \times 2.5 \times 1.51 = −18.5$	$−12.1 \times 2.5 \times 1.51 = −45.7$	4.9	−48.7	0.94
Sidewalls	−8.3	−15.5	$−8.3 \times 2.5 \times 1.51 = −31.4$	$−15.5 \times 2.5 \times 1.51 = −58.6$	−3.9	−57.5	1.02
MWFRS: Roof							
Wind perpendicular to 30-ft side (largest pressures)	−11.7	−18.9	$−12.8 \times 2.5 \times 1.51 = −44.2$	$−18.9 \times 2.5 \times 1.51 = −71.4$	−12.7	−66.2	1.08
Wind perpendicular to 60-ft side (largest pressures)	−18.5	−25.7	$−18.5 \times 2.5 \times 1.51 = −69.9$	$−25.7 \times 2.5 \times 1.51 = −97.1$	−30.2	−83.7	1.16

walls, additional shear walls in the interior of the building may reduce the tendency for the building to rack and/or overturn. Finally, adequate shear wall design requires proper anchorage at the ends of the shear walls and sometimes at the ends of shear wall segments. This anchorage is inherent in some types of

structural systems such as reinforced masonry that follows the latest design requirements but must be explicitly added for other types of structural systems such as wood frame.

If it is desired to avoid collapse of interior walls and/or ceilings in the event that exterior glazing is breached, loads on these interior

elements need to be calculated. Resistance to failure may require additional or strengthened connections between the top of interior walls and the roof or ceiling systems and connections between the bottom of interior walls and the floor and foundation systems. Reasonable design loads for use in designing these elements and their connections would be 80% of the exterior wall design loads.

Exterior Glazing. Glazing damage is prevalent during tornadoes (Rouche and Prevatt 2013). To minimize breaching of exterior glazing by EF0–EF2 rated debris, specify glazing assemblies that have been designed and tested in accordance with ASTM E1886, using ASTM E1996 test missile D or preferably E. To avoid breaching of exterior glazing by EF3–EF5 rated debris, specify glazing assemblies that have been tested in accordance with AAMA 512, using test missiles given in ICC 500) or FEMA P-361 (FEMA 2015).

Critical Facilities. For design guidance to minimize building damage to critical facilities, see Tornado Recovery Advisory No. 6, *Critical Facilities Located in Tornado-Prone Regions: Recommendations for Architects and Engineers* (FEMA 2011) and FEMA P-908 (FEMA 2012). This Tornado Recovery Advisory includes detailed recommendations for three levels of enhancement for the MWFRS and C&C to minimize building damage.

C26.14.5 Continuity of Building Operations. Designing a building to ensure that it will remain operational if struck by an EF4 or EF5 rated tornado is expensive. However, for those facilities such as emergency operations centers and hospitals where it is desired to avoid interrupted operations, see FEMA (2011) in FEMA P-908 (FEMA 2012). This Tornado Recovery Advisory includes detailed recommendations related to the MWFRS, the building envelope, HVAC, water, sewer, and emergency power.

C26.14.6 Trussed Communications Towers. Damage investigations have shown that wind-borne debris can cling to trussed communications towers (FEMA 2012). Clinging debris such as metal roof panels and chain-link fence with privacy slats can increase A_f , the projected surface area subject to wind loading, thus increasing the net wind load on the structure, with tower collapse as a possible result. To minimize the collapse potential, towers can be designed for the additional wind load caused by clinging debris.

When it is desired to design for clinging debris, consider the potential debris sources in the vicinity of the tower. Based on FEMA (2012) damage observations, as a minimum, design for 40 ft² (3.7 m²) of projected surface area of clinging debris located at midheight of the tower or 50 ft (15 m), whichever is lower.

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